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Name of Faculty: Dr. Raish Muhammad

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UNIT-V (VECTOR CALCULUS)

LECTURE NO.- 32

Introduction

In vector calculus, we deal with two types of functions: Scalar Functions (or Scalar Fields) and Vector Functions (or Vector Fields) and their differentiation and integration.

Scalar Fields

A scalar quantity possesses only magnitude and has no concern with direction. A single real number gives us a complete representation of a scalar quantity. Thus a scalar quantity is nothing but a real number.

Let D be any subset of the set of all real numbers. If to each element t of D , we associate by some rule, a unique real number $f(t)$, then this rule defines a **scalar function** of the scalar variable t . Here $f(t)$ is a scalar quantity and thus f is a scalar function.

If to each point $P(x, y, z)$ of a region R in space, there corresponds a unique scalar $f(P) = f(x, y, z)$, then f is called a **scalar point function** and the set of all scalars $f(P)$ for all values of (x, y, z) in R is called a **scalar field** f over region R .

Precisely, we can say that scalar function defines a scalar field in a region or on a space or a curve. Examples are the temperature field in a body, pressure field in the air in earth's atmosphere.

Moreover, if the position vector of the point P is $\vec{r}$, then we may also write the scalar field as $F(P) = F(\vec{r})$. This notation emphasizes the fact that the scalar value $F(\vec{r})$ is associated with the position vector $\vec{r}$ in the region R .

Examples of scalar fields:

E.g. 1) The distance $F(P)$ of any point $P(x, y, z)$ from a fixed point $P'(x', y', z')$ in the space is a scalar function whose domain of definition is the whole space and is given by $F(P) = \sqrt{(x - x')^2 + (y - y')^2 + (z - z')^2}$. Also $F(P)$ defines a scalar field in space.

E.g. 2) The function $F(x, y, z) = xy^2 + yz + x^2$ for the point (x, y, z) inside the unit sphere $x^2 + y^2 + z^2 = 1$ is a scalar function and also defines a scalar field throughout the sphere.

Ex. 3) The temperature at any point within or on the surface of earth at a certain time defines a scalar field.

Ex. 4) The gravitational potential of a system of masses in a medium defines a scalar field.

Note: In the physical problems, the scalar function F depends on time variable t in addition to the point P and then we write it as $F(P, t) = F(\vec{r}, t) = F(x, y, z, t)$. The example of such a time dependent scalar function is the temperature distribution throughout a block of metal heated in such a way that its temperature varies with time.

Vector Fields

Let D be any subset of the set of all real numbers. If to each element t of D , we associate by some rule, a unique vector $\vec{f}(t)$, then this rule defines a vector function of the scalar variable t . Here $\vec{f}(t)$ is a vector quantity and thus $\vec{f}$ is a vector function.

If to each point $P(x, y, z)$ of a region R in space there corresponds a unique vector $\vec{f}(P) = \vec{f}(x, y, z)$, then $\vec{f}$ is called a **vector point function** and the set of all vectors $\vec{f}(P)$ for all points $P(x, y, z)$ in R is called a **vector field** $\vec{f}$ over region R .

Moreover, if the position vector of the point P is $\vec{r}$, then we may write the vector field as $\vec{f}(P) = \vec{f}(\vec{r})$. This notation emphasizes the fact that the vector value $\vec{f}(\vec{r})$ is associated with the position vector $\vec{r}$ in the region R . Also the general form (component form) of the vector function is $\vec{f}(\vec{r}) = F_1(\vec{r})\hat{i} + F_2(\vec{r})\hat{j} + F_3(\vec{r})\hat{k}$, where the components $F_1(\vec{r})$, $F_2(\vec{r})$ and $F_3(\vec{r})$ are the scalar functions.

Examples of vector fields:

E.g. 1) The function $\vec{F}(x, y, z) = 2xy\hat{i} + \sin x\hat{j} + 3z^2\hat{k}$ for point $P(x, y, z)$ inside an ellipsoid $\frac{x^2}{9} + \frac{y^2}{16} + \frac{z^2}{4} = 1$ is a vector function and defines a vector field throughout the ellipsoid.

E.g. 2) The force field given by $\vec{F}(x, y, z) = x\hat{i} + 2y\hat{j} + z^2\hat{k}$ is a vector field.

Ex. 3) The velocity at any point (x, y, z) of a particle moving in a curve at a certain time defines a vector field.

Ex. 4) The acceleration of a moving particle is a vector field.

Note: Like the time dependent scalar field, time dependent vector field also exists. Such a field depends on time variable t in addition to the point in the region R and may be expressed as $\vec{F}(\vec{r}, t) = F_1(\vec{r}, t)\hat{i} + F_2(\vec{r}, t)\hat{j} + F_3(\vec{r}, t)\hat{k}$, where F_1 , F_2 and F_3 are scalar functions. An example of time dependent vector field is the fluid velocity vector in the unsteady flow of water around a bridge support column, because this velocity depends on the position vector $\vec{r}$ in the water and the time variable t and is given as $\vec{V}(\vec{r}, t)$.

Vector Functions of a Single Variable

A vector function $\vec{F}$ of single variable t is a function which assigns a vector value $\vec{F}(t)$ to each scalar value t in interval $a \leq t \leq b$. In the component form, it may be written as $\vec{F}(t) = F_1(t)\hat{i} + F_2(t)\hat{j} + F_3(t)\hat{k}$ where F_1 , F_2 and F_3 are called components and are scalar functions of the same single variable t .

For example, the functions given by $\vec{F}(t) = t\hat{i} + \sin(t-2)\hat{j} + \cos 3t\hat{k}$ and $\vec{G}(t) = t^2\hat{i} + e^t\hat{j} + \log t\hat{k}$ are vector functions of a single variable t .

Limit of a Vector Function of a Single Variable

A vector function $\vec{F}(t) = F_1(t)\hat{i} + F_2(t)\hat{j} + F_3(t)\hat{k}$ of single variable t is said to have a limit $\vec{L} = L_1\hat{i} + L_2\hat{j} + L_3\hat{k}$ as $t \rightarrow t_0$, if $\vec{F}(t)$ is defined in the neighborhood of t_0 and $\lim_{t \rightarrow t_0} |\vec{F}(t) - \vec{L}| = 0$ or $\lim_{t \rightarrow t_0} |F_1(t) - L_1| = \lim_{t \rightarrow t_0} |F_2(t) - L_2| = \lim_{t \rightarrow t_0} |F_3(t) - L_3| = 0$, then we write it as $\lim_{t \rightarrow t_0} \vec{F}(t) = \vec{L}$.

Continuity of a Vector Function of a Single Variable

A vector function $\vec{F}(t) = F_1(t)\hat{i} + F_2(t)\hat{j} + F_3(t)\hat{k}$ of a single variable t is said to be continuous at $t = t_0$, if it is defined in some neighborhood of t_0 and $\lim_{t \rightarrow t_0} \vec{F}(t) = \vec{F}(t_0)$.

Moreover, $\vec{F}(t)$ is said to be continuous at $t = t_0$ if and only if its three components F_1, F_2 and F_3 are continuous as $t = t_0$.

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Differentiation of Vector Functions:

Let $\vec{F}(t)$ be a vector function of the scalar variable t . Then if exists, the derivative of $\vec{F}(t)$ with respect to t is denoted by $\frac{d\vec{F}}{dt}$ and defined as

$$\frac{d\vec{F}}{dt} = \lim_{\delta t \rightarrow 0} \frac{\vec{F}(t+\delta t) - \vec{F}(t)}{(t+\delta t) - t} = \lim_{\delta t \rightarrow 0} \frac{\delta \vec{F}(t)}{\delta t}.$$

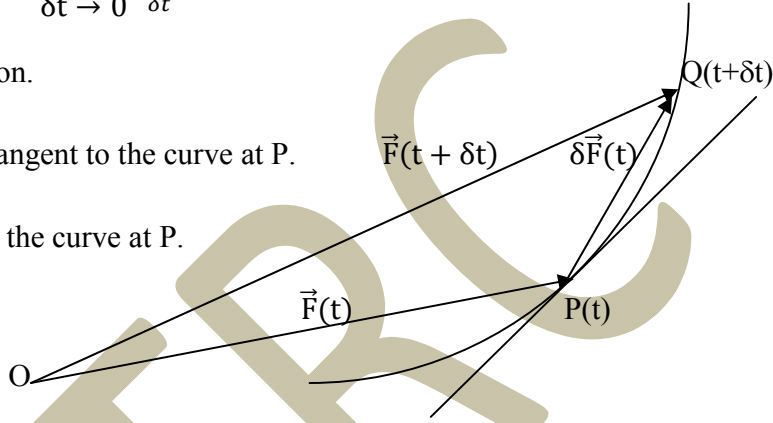
$\frac{d\vec{F}}{dt}$ is also a vector function.

As $Q \rightarrow P$, PQ becomes tangent to the curve at P .

So $\frac{d\vec{F}}{dt}$ is along tangent to the curve at P .

Now its magnitude

$$\begin{aligned} \left| \frac{d\vec{F}}{dt} \right| &= \lim_{\delta t \rightarrow 0} \frac{|\delta \vec{F}(t)|}{\delta t} = \lim_{\delta t \rightarrow 0} \frac{|\delta \vec{F}(t)|}{\delta s} \cdot \frac{\delta s}{\delta t}, \quad |\delta \vec{F}(t)| \text{ is chord } PQ \text{ and } \delta s \text{ is arc } PQ \\ &= \lim_{\delta t \rightarrow 0} \frac{|\delta \vec{F}(t)|}{\delta s} \cdot \lim_{\delta t \rightarrow 0} \frac{\delta s}{\delta t} = 1 \cdot \lim_{\delta t \rightarrow 0} \frac{\delta s}{\delta t}, \text{ since chord } PQ = \text{arc } PQ \text{ as } Q \rightarrow P \\ &= \frac{ds}{dt}. \end{aligned}$$

**Properties of Differentiation of the Vector Functions:**

Let $\vec{F}(t)$, $\vec{G}(t)$ and $\vec{H}(t)$ be the differentiable vector functions of a scalar variable t and $\phi(t)$ be a differentiable scalar function of t , then

- (i) $\frac{d}{dt} (\vec{F} \pm \vec{G}) = \frac{d\vec{F}}{dt} \pm \frac{d\vec{G}}{dt}$
- (ii) $\frac{d}{dt} (\vec{F} \cdot \vec{G}) = \vec{F} \cdot \frac{d\vec{G}}{dt} + \vec{G} \cdot \frac{d\vec{F}}{dt}$
- (iii) $\frac{d}{dt} (\vec{F} \times \vec{G}) = \vec{F} \times \frac{d\vec{G}}{dt} + \frac{d\vec{F}}{dt} \times \vec{G}$
- (iv) $\frac{d}{dt} (\phi \vec{F}) = \phi \frac{d\vec{F}}{dt} + \vec{F} \frac{d\phi}{dt}$
- (v) $\frac{d}{dt} \begin{bmatrix} \vec{F} & \vec{G} & \vec{H} \end{bmatrix} = \begin{bmatrix} \frac{d\vec{F}}{dt} & \vec{G} & \vec{H} \end{bmatrix} + \begin{bmatrix} \vec{F} & \frac{d\vec{G}}{dt} & \vec{H} \end{bmatrix} + \begin{bmatrix} \vec{F} & \vec{G} & \frac{d\vec{H}}{dt} \end{bmatrix}$
- (vi) $\frac{d}{dt} \{ \vec{F} \times (\vec{G} \times \vec{H}) \} = \frac{d\vec{F}}{dt} \times (\vec{G} \times \vec{H}) + \vec{F} \times \left(\frac{d\vec{G}}{dt} \times \vec{H} \right) + \vec{F} \times \left(\vec{G} \times \frac{d\vec{H}}{dt} \right)$

Derivative of a Constant Vector: $\frac{d\vec{C}}{dt} = \vec{0}$.

Derivative of vector function in terms of its components:

Let $\vec{F}(t)$ be a vector function of a scalar variable t such that

$$\vec{F}(t) = f_1(t)\hat{i} + f_2(t)\hat{j} + f_3(t)\hat{k}, \text{ then } \frac{d\vec{F}}{dt} = \frac{df_1}{dt}\hat{i} + \frac{df_2}{dt}\hat{j} + \frac{df_3}{dt}\hat{k}.$$

Successive Differentiation:

If $\frac{d\vec{F}}{dt}$ exists, then the vector function $\vec{F}(t)$ is called differentiable. If we again differentiate $\frac{d\vec{F}}{dt}$ with respect to t , then it is denoted by $\frac{d^2\vec{F}}{dt^2}$ and is called second derivative of $\vec{F}(t)$ w.r.t. t and so on. We can continue differentiating successively up to n times and get $\frac{d^n\vec{F}}{dt^n}$.

Examples on Differentiation of Vector Functions:

Example 1. Show that if $\vec{r} = \vec{a} \sin \omega t + \vec{b} \cos \omega t$ where $\vec{a}$, $\vec{b}$ and ω are constants, then

$$\frac{d^2\vec{r}}{dt^2} = -\omega^2\vec{r} \text{ and } \vec{r} \times \frac{d\vec{r}}{dt} = -\omega(\vec{a} \times \vec{b}).$$

Solution: Given $\vec{r} = \vec{a} \sin \omega t + \vec{b} \cos \omega t$

$$\text{Differentiating w. r. to } t, \quad \frac{d\vec{r}}{dt} = \vec{a}\omega \cos \omega t - \vec{b}\omega \sin \omega t$$

$$\begin{aligned} \text{Again differentiating w. r. to } t, \quad \frac{d^2\vec{r}}{dt^2} &= -\vec{a}\omega^2 \sin \omega t - \vec{b}\omega^2 \cos \omega t \\ &= -\omega^2(\vec{a} \sin \omega t + \vec{b} \cos \omega t) = -\omega^2\vec{r} \end{aligned}$$

$$\begin{aligned} \text{Also } \vec{r} \times \frac{d\vec{r}}{dt} &= (\vec{a} \sin \omega t + \vec{b} \cos \omega t) \times (\vec{a}\omega \cos \omega t - \vec{b}\omega \sin \omega t) \\ &= (\vec{a} \times \vec{a})\omega \sin \omega t \cos \omega t + (\vec{b} \times \vec{a})\omega \cos^2 \omega t - (\vec{a} \times \vec{b})\omega \sin^2 \omega t - (\vec{b} \times \vec{b})\omega \sin \omega t \cos \omega t \\ &= -(\vec{a} \times \vec{b})\omega(\cos^2 \omega t + \sin^2 \omega t) \quad (\text{since } \vec{a} \times \vec{a} = \vec{b} \times \vec{b} = \vec{0}) \\ &= -(\vec{a} \times \vec{b})\omega = -\omega(\vec{a} \times \vec{b}). \end{aligned}$$

Example 2. If $\vec{P} = 5t^2\hat{i} + t^3\hat{j} - t\hat{k}$ and $\vec{Q} = 2 \sin t \hat{i} - \cos t \hat{j} + 5t\hat{k}$, then find

$$(a) \frac{d}{dt} (\vec{P} \cdot \vec{Q}) \text{ and } (b) \frac{d}{dt} (\vec{P} \times \vec{Q}).$$

Solution: Consider $\vec{P} = 5t^2\hat{i} + t^3\hat{j} - t\hat{k}$ and $\vec{Q} = 2 \sin t \hat{i} - \cos t \hat{j} + 5t\hat{k}$

$$\text{So } \frac{d\vec{P}}{dt} = 10t\hat{i} + 3t^2\hat{j} - \hat{k} \text{ and } \frac{d\vec{Q}}{dt} = 2 \cos t \hat{i} + \sin t \hat{j} + 5\hat{k}$$

$$\begin{aligned} a) \quad \frac{d}{dt} (\vec{P} \cdot \vec{Q}) &= \frac{d\vec{P}}{dt} \cdot \vec{Q} + \vec{P} \cdot \frac{d\vec{Q}}{dt} \\ &= (10t\hat{i} + 3t^2\hat{j} - \hat{k}) \cdot (2 \sin t \hat{i} - \cos t \hat{j} + 5t\hat{k}) \\ &\quad + (5t^2\hat{i} + t^3\hat{j} - t\hat{k}) \cdot (2 \cos t \hat{i} + \sin t \hat{j} + 5\hat{k}) \end{aligned}$$

$$= 20t \sin t - 3t^2 \cos t - 5t + 10t^2 \cos t + t^3 \sin t - 5t$$

$$= t^3 \sin t + 7t^2 \cos t + 20t \sin t - 10t$$

$$\begin{aligned} \text{b) } \frac{d}{dt} (\vec{P} \times \vec{Q}) &= \frac{d\vec{P}}{dt} \times \vec{Q} + \vec{P} \times \frac{d\vec{Q}}{dt} \\ &= (10t \hat{i} + 3t^2 \hat{j} - \hat{k}) \times (2 \sin t \hat{i} - \cos t \hat{j} + 5t \hat{k}) \\ &\quad + (5t^2 \hat{i} + t^3 \hat{j} - t \hat{k}) \times (2 \cos t \hat{i} + \sin t \hat{j} + 5 \hat{k}) \\ &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 10t & 3t^2 & -1 \\ 2 \sin t & -\cos t & 5t \end{vmatrix} + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 5t^2 & t^3 & -t \\ 2 \cos t & \sin t & 5 \end{vmatrix} \\ &= \{ \hat{i}(15t^3 - \cos t) + \hat{j}(-2 \sin t - 50t^2) + \hat{k}(-10t \cos t - 6t^2 \sin t) \} \\ &\quad + \{ \hat{i}(5t^3 + t \sin t) + \hat{j}(-2t \cos t - 25t^2) + \hat{k}(5t^2 \sin t - 2t^3 \cos t) \} \\ &= \hat{i}(20t^3 + t \sin t - \cos t) - \hat{j}(2t \cos t + 2 \sin t + 75t^2) \\ &\quad - \hat{k}(2t^3 \cos t + 10t \cos t + t^2 \sin t) \end{aligned}$$

Example 3. If $\vec{a} = x^2yz \hat{i} - 2xz^3 \hat{j} + xz^2 \hat{k}$ and $\vec{b} = 2z \hat{i} + y \hat{j} - x^2 \hat{k}$, find $\frac{\partial^2}{\partial x \partial y} (\vec{a} \times \vec{b})$ at $(1, 0, -2)$.

Solution: Here $\vec{a} \times \vec{b} = (x^2yz \hat{i} - 2xz^3 \hat{j} + xz^2 \hat{k}) \times (2z \hat{i} + y \hat{j} - x^2 \hat{k})$

$$= x^2y^2z \hat{k} + x^4yz \hat{j} + 4xz^4 \hat{k} + 2x^3z^3 \hat{i} + 2xz^3 \hat{j} - xyz^2 \hat{i}$$

$$(\because \hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = \vec{0} \text{ and } \hat{i} \times \hat{j} = \hat{k}, \hat{j} \times \hat{i} = -\hat{k} \text{ etc.})$$

$$= (2x^3z^3 - xyz^2) \hat{i} + (x^4yz + 2xz^3) \hat{j} + (x^2y^2z + 4xz^4) \hat{k}$$

$$\begin{aligned} \text{Now } \frac{\partial^2}{\partial x \partial y} (\vec{a} \times \vec{b}) &= \frac{\partial^2}{\partial x \partial y} [(2x^3z^3 - xyz^2) \hat{i} + (x^4yz + 2xz^3) \hat{j} + (x^2y^2z + 4xz^4) \hat{k}] \\ &= \frac{\partial}{\partial x} \frac{\partial}{\partial y} [(2x^3z^3 - xyz^2) \hat{i} + (x^4yz + 2xz^3) \hat{j} + (x^2y^2z + 4xz^4) \hat{k}] \\ &= \frac{\partial}{\partial x} [-xz^2 \hat{i} + x^4z \hat{j} + 2x^2yz \hat{k}] = [-z^2 \hat{i} + 4x^3z \hat{j} + 4xyz \hat{k}] \end{aligned}$$

$$\text{At the point } (1, 0, -2) \quad \frac{\partial^2}{\partial x \partial y} (\vec{a} \times \vec{b}) = -4\hat{i} - 8\hat{j}$$

Example 4. If $\hat{a}$ be a variable unit vector, show that $\frac{d\hat{a}}{dt}$ is a vector normal to $\hat{a}$ and $\frac{d\hat{a}}{d\theta}$ is a unit vector normal to $\hat{a}$, where θ is the angle through which $\hat{a}$ turns.

Solution. Try yourself.

Example 5. Derive radial and transverse velocities and accelerations of a particle describing a plane curve, with the help of vectors.

Solution. Try yourself.

Example 6. Find by vector methods, the tangential and normal accelerations of a particle moving in a plane curve.

Solution. Try yourself.

Example 7. A particle moves on the curve $x = 2t^2, y = t^2 - 4t, z = 3t - 5$, where t is the time variable. Find the components of velocity and acceleration at $t = 1$ in the direction $\hat{i} - 3\hat{j} + 2\hat{k}$.

Solution. Try yourself.

Example 8. Find the unit tangent vector at any point on the curve

$$x = t^2 + 1, y = 4t - 3, z = 2t^2 - 6t,$$

where t is any variable. Also determine the unit tangent vector at the point $t = 2$.

Solution. Try yourself.

Integration of Vector Functions:

Integration is the reverse process of the differentiation. If $\vec{F}(t)$ and $\vec{G}(t)$ be two vector functions of the scalar t such that $\frac{d\vec{G}}{dt} = \vec{F}(t)$, then

$$\int \vec{F}(t) dt = \vec{G}(t) + \vec{C},$$

where $\vec{C}$ is an arbitrary constant vector.

$\int \vec{F}(t) dt$ is called the indefinite integral of the vector function $\vec{F}(t)$ w.r.t. t .

Then the vector function $\vec{F}(t)$, which is being integrated is called the integrand.

If the integration is defined over the closed interval $[a, b]$, then the definite integral between the limits $t = a$ and $t = b$ is defined as

$$\int_a^b \vec{F}(t) dt = \vec{G}(b) - \vec{G}(a),$$

where a and b are called the limits of integration.

Examples on Integration of Vector Functions:

Example 9. If $\frac{d^2\vec{f}}{dt^2} = 6t\hat{i} - 12t^2\hat{j} + 4\cos t\hat{k}$, find $\vec{f}$, given that $\frac{d\vec{f}}{dt} = -\hat{i} - 3\hat{k}$ and $\vec{f} = 2\hat{i} + \hat{j}$ at $t = 0$.

Solution: Given that $\frac{d^2\vec{f}}{dt^2} = 6t\hat{i} - 12t^2\hat{j} + 4\cos t\hat{k}$... (1)

Integrating (1) with respect to t ,

$$\int \frac{d^2\vec{f}}{dt^2} dt = \int (6t\hat{i} - 12t^2\hat{j} + 4\cos t\hat{k}) dt$$

Implying $\frac{d\vec{f}}{dt} = 3t^2\hat{i} - 4t^3\hat{j} + 4\sin t\hat{k} + \vec{c}_1$... (2)

Now integrating (2) w.r.t. t , we get

$$\int \frac{d\vec{f}}{dt} dt = \int (3t^2 \hat{i} - 4t^3 \hat{j} + 4 \sin t \hat{k} + \vec{c}_1) dt$$

$$\text{Implying } \vec{f} = t^3 \hat{i} - t^4 \hat{j} - 4 \cos t \hat{k} + \vec{c}_1 t + \vec{c}_2 \quad \dots (3)$$

$$\text{Also we are given that at } t = 0, \quad \frac{d\vec{f}}{dt} = -\hat{i} - 3\hat{k} \quad \dots (4)$$

$$\vec{f} = 2\hat{i} + \hat{j} \quad \dots (5)$$

$$\text{Using (2) and (4), } \vec{c}_1 = -\hat{i} - 3\hat{k}$$

$$\text{Using (3) and (5), } -4\hat{k} + \vec{c}_2 = 2\hat{i} + \hat{j} \Rightarrow \vec{c}_2 = 2\hat{i} + \hat{j} + 4\hat{k}$$

Putting the values of the constant vectors $\vec{c}_1$ & $\vec{c}_2$ in (3), we get

$$\begin{aligned} \vec{f} &= t^3 \hat{i} - t^4 \hat{j} - 4 \cos t \hat{k} + (-\hat{i} - 3\hat{k})t + (2\hat{i} + \hat{j} + 4\hat{k}) \\ &= (t^3 - t + 2)\hat{i} + (1 - t^4)\hat{j} - (4 \cos t + 3t + 4)\hat{k} \end{aligned}$$

Example 10. The acceleration of a particle at any time $t \geq 0$ is given by

$$\vec{a} = 12 \cos 2t \hat{i} - 8 \sin 2t \hat{j} + 16t \hat{k}$$

If the velocity $\vec{v}$ and displacement $\vec{r}$ are zero at $t = 0$, find $\vec{v}$ and $\vec{r}$ at any time.

Example 11. For given $\vec{f}(t) = (3t^2 - 2t)\hat{i} + (6t - 4)\hat{j} + 4t\hat{k}$, evaluate $\int_2^3 \vec{f}(t) dt$.

Solution. Try yourself.

Example 12. If $\vec{r}(t) = 5t^2\hat{i} + t\hat{j} - t^3\hat{k}$, evaluate $\int_1^2 \left(\vec{r} \times \frac{d^2\vec{r}}{dt^2} \right) dt$.

Solution. Try yourself.

Example 13. Given that $\vec{r} = 2\hat{i} - \hat{j} + 2\hat{k}$, when $t = 2$ and $\vec{r} = 4\hat{i} - 2\hat{j} + 3\hat{k}$, when $t = 3$,

show that $\int_2^3 \left(\vec{r} \cdot \frac{d\vec{r}}{dt} \right) dt = 10$.

Solution. Try yourself.

LECTURE NO.:- 34

Vector Differential Operator ∇ (Del)

The vector differential operator ∇ (read as del or nabla) is defined as

$$\nabla \equiv \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \equiv \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \text{ and operates distributively.}$$

The del operator (∇) can generally be treated to behave as an ordinary vector. It possesses properties like ordinary vectors. The symbols $\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}$ are partial differential operators and can be treated as its components along $\hat{i}, \hat{j}, \hat{k}$.

Gradient of a Scalar Field

Let $f(x, y, z)$ be defined and differentiable at each point (x, y, z) in a certain region of space i.e. defines a differential scalar field. Then the gradient of f , written as ∇f or $\text{grad } f$, is defined as

$$\nabla f = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) f = \hat{i} \frac{\partial f}{\partial x} + \hat{j} \frac{\partial f}{\partial y} + \hat{k} \frac{\partial f}{\partial z} = \sum \hat{i} \frac{\partial f}{\partial x}.$$

It should be noted that ∇f is a vector whose three components are $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}$. Thus the gradient of a scalar field defines a vector field. If f is a scalar point function, then ∇f is a vector point function.

Some Important Results involving Gradient:

- Gradient of the sum of two scalar point functions:** If f and g are two scalar point functions, then $\text{grad } (f + g) = \text{grad } f + \text{grad } g$ or $\nabla(f + g) = \nabla f + \nabla g$.
- Gradient of a constant scalar:** If c be a constant scalar, then $\nabla c = \vec{0}$.
- Gradient of the product of two scalar point function:** If f and g be two scalar point functions, then $\text{grad } (fg) = f \text{ grad } g + g \text{ grad } f$ or $\nabla(fg) = f \nabla g + g \nabla f$.
- Gradient of quotient of two scalar functions:** If f and g are two scalar point functions, then $\text{grad} \left(\frac{f}{g} \right) = \frac{g(\text{grad } f) - f(\text{grad } g)}{g^2}$ or $\nabla \left(\frac{f}{g} \right) = \frac{g(\nabla f) - f(\nabla g)}{g^2}$.
- Gradient of the scalar multiplication of scalar point function:** If f is a scalar point function and c be a constant scalar, then $\nabla(cf) = c \nabla f$.

Divergence of a Vector Point Function

Let $\vec{V}$ be any differentiable vector point function. Then the divergence of $\vec{V}$, written as $\nabla \cdot \vec{V}$ or $\text{div } \vec{V}$, is defined as

$$\text{div } \vec{V} = \nabla \cdot \vec{V} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot \vec{V} = \hat{i} \cdot \frac{\partial \vec{V}}{\partial x} + \hat{j} \cdot \frac{\partial \vec{V}}{\partial y} + \hat{k} \cdot \frac{\partial \vec{V}}{\partial z} = \sum \hat{i} \cdot \frac{\partial \vec{V}}{\partial x}.$$

It should be noted that $\text{div } \vec{V}$ is a scalar quantity. Thus the divergence of a vector point function is a scalar point function.

Note: If $\vec{V} = V_1\hat{i} + V_2\hat{j} + V_3\hat{k}$ is a differentiable vector point function, then

$$\text{div } \vec{V} = \nabla \cdot \vec{V} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot (V_1\hat{i} + V_2\hat{j} + V_3\hat{k}) = \frac{\partial V_1}{\partial x} + \frac{\partial V_2}{\partial y} + \frac{\partial V_3}{\partial z}.$$

Solenoidal Vector: A vector $\vec{V}$ is said to be solenoidal, if $\text{div } \vec{V} = 0$.

Some Important Results involving Divergence:

- 1. Divergence of the sum of two vector point functions:** If $\vec{f}$ and $\vec{g}$ are two scalar point functions, then $\text{div } (\vec{f} + \vec{g}) = \text{div } \vec{f} + \text{div } \vec{g}$ or $\nabla \cdot (\vec{f} + \vec{g}) = \nabla \cdot \vec{f} + \nabla \cdot \vec{g}$.
- 2. Divergence of a constant vector:** If $\vec{c}$ be a constant vector, then $\nabla \cdot \vec{c} = 0$.
- 3. Divergence of the scalar multiplication of vector point function:** If $\vec{f}$ is a vector point function and c be a constant scalar, then $\nabla \cdot (c\vec{f}) = c \nabla \cdot \vec{f}$.

Curl of a Vector Point Function

Let $\vec{V}$ be a given differentiable vector point function. Then the curl or rotation of $\vec{V}$, written as $\nabla \times \vec{V}$ or $\text{curl } \vec{V}$ or $\text{rot } \vec{V}$ is defined as

$$\text{curl } \vec{V} = \nabla \times \vec{V} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \times \vec{V} = \hat{i} \times \frac{\partial \vec{V}}{\partial x} + \hat{j} \times \frac{\partial \vec{V}}{\partial y} + \hat{k} \times \frac{\partial \vec{V}}{\partial z} = \sum \hat{i} \times \frac{\partial \vec{V}}{\partial x}.$$

It should be noted that $\text{curl } \vec{V}$ is a vector quantity. Thus the curl of a vector point function is a vector point function.

If $\vec{V}$ be the linear velocity at a point, then $\nabla \times \vec{V} = \vec{\omega}$ be the angular velocity at that point.

Note. If $\vec{V} = V_1\hat{i} + V_2\hat{j} + V_3\hat{k}$ is a differentiable vector point function, then

$$\begin{aligned} \text{curl } \vec{V} &= \nabla \times \vec{V} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \times (V_1\hat{i} + V_2\hat{j} + V_3\hat{k}) \\ &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ V_1 & V_2 & V_3 \end{vmatrix} \\ &= \hat{i} \left(\frac{\partial V_3}{\partial y} - \frac{\partial V_2}{\partial z} \right) + \hat{j} \left(\frac{\partial V_1}{\partial z} - \frac{\partial V_3}{\partial x} \right) + \hat{k} \left(\frac{\partial V_2}{\partial x} - \frac{\partial V_1}{\partial y} \right) = \sum \hat{i} \left(\frac{\partial V_3}{\partial y} - \frac{\partial V_2}{\partial z} \right). \end{aligned}$$

Irrotational Vector: A vector $\vec{V}$ is said to be irrotational, if $\text{curl } \vec{V} = \nabla \times \vec{V} = \vec{0}$.

Scalar Potential of an Irrotational Vector Field: If a vector field $\vec{V}$ is irrotational, then a scalar point function, say ϕ exists such that $\vec{V} = \nabla \phi$, where ϕ is called a scalar potential of $\vec{V}$.

In this case $\vec{V}$ is also called a conservative vector field or a vector field of potential kind.

Some Important Results involving Curl:

- 1. Curl of the sum of two vector point functions:** If $\vec{f}$ and $\vec{g}$ are two scalar point functions, then $\text{curl}(\vec{f} + \vec{g}) = \text{curl} \vec{f} + \text{curl} \vec{g}$ or $\nabla \times (\vec{f} + \vec{g}) = \nabla \times \vec{f} + \nabla \times \vec{g}$.
- 2. Curl of a constant vector:** If $\vec{c}$ be a constant vector, then $\nabla \times \vec{c} = \vec{0}$.
- 3. Curl of the scalar multiplication of vector point function:** If $\vec{f}$ is a vector point function and c be a constant scalar, then $\nabla \times (c \vec{f}) = c \nabla \times \vec{f}$.

Laplacian Operator ∇^2

The Laplacian operator ∇^2 is defined as $\nabla^2 \equiv \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$.

If f is a scalar point function, then $\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = \nabla \cdot \nabla f = \text{div}(\text{grad } f)$.

It should be noted that $\nabla^2 f$ is also a scalar quantity.

If $\vec{f}$ is a vector point function, then $\nabla^2 \vec{f} = \frac{\partial^2 \vec{f}}{\partial x^2} + \frac{\partial^2 \vec{f}}{\partial y^2} + \frac{\partial^2 \vec{f}}{\partial z^2}$.

It should be noted that $\nabla^2 \vec{f}$ is also a vector quantity.

Laplace's Equation: The equation $\nabla^2 f = 0$ is called Laplace's equation. The function f which satisfies Laplace's equation is called a harmonic function.

Some Important Vector Identities

If $\vec{A}$, $\vec{B}$ are differentiable vector functions and ϕ is a differentiable scalar function, then

- $\text{div}(\phi \vec{A}) = (\text{grad } \phi) \cdot \vec{A} + \phi \text{div} \vec{A}$ or $\nabla \cdot (\phi \vec{A}) = (\nabla \phi) \cdot \vec{A} + \phi (\nabla \cdot \vec{A})$.
- $\text{curl}(\phi \vec{A}) = (\text{grad } \phi) \times \vec{A} + \phi \text{curl} \vec{A}$ or $\nabla \times (\phi \vec{A}) = (\nabla \phi) \times \vec{A} + \phi (\nabla \times \vec{A})$.
- $\text{div}(\vec{A} \times \vec{B}) = \vec{B} \cdot \text{curl} \vec{A} - \vec{A} \cdot \text{curl} \vec{B}$ or $\nabla \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\nabla \times \vec{A}) - \vec{A} \cdot (\nabla \times \vec{B})$.
- $\text{curl}(\vec{A} \times \vec{B}) = (\vec{B} \cdot \nabla) \vec{A} - \vec{B} \text{div} \vec{A} - (\vec{A} \cdot \nabla) \vec{B} + \vec{A} \text{div} \vec{B}$.
- $\text{grad}(\vec{A} \cdot \vec{B}) = (\vec{B} \cdot \nabla) \vec{A} + (\vec{A} \cdot \nabla) \vec{B} + \vec{B} \times \text{curl} \vec{A} + \vec{A} \times \text{curl} \vec{B}$.
- $\text{div grad } \phi = \nabla^2 \phi$ i.e., $\nabla \cdot (\nabla \phi) = \nabla^2 \phi$.
- $\text{div curl } \vec{A} = 0$ i.e., $\nabla \cdot (\nabla \times \vec{A}) = 0$.
- $\nabla \times (\nabla \times \vec{A}) = \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A}$.
- curl of the gradient of ϕ is zero i.e., $\nabla \times (\nabla \phi) = \vec{0}$.

Note. If a vector field $\vec{V}$ is irrotational, then $\nabla \times \vec{V} = \vec{0}$. Using the identity $\nabla \times (\nabla \phi) = \vec{0}$, there exists a scalar function ϕ corresponding to the irrotational vector field $\vec{V}$ such that $\vec{V} = \nabla \phi$. In this case, ϕ is called a scalar potential function of $\vec{V}$.

$$\text{If } \vec{V} = V_1 \hat{i} + V_2 \hat{j} + V_3 \hat{k}, \text{ then } \nabla \phi = \hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y} + \hat{k} \frac{\partial \phi}{\partial z} = V_1 \hat{i} + V_2 \hat{j} + V_3 \hat{k}.$$

$$\text{It gives } \frac{\partial \phi}{\partial x} = V_1, \frac{\partial \phi}{\partial y} = V_2, \frac{\partial \phi}{\partial z} = V_3.$$

$$\therefore d\phi = \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy + \frac{\partial \phi}{\partial z} dz = V_1 dx + V_2 dy + V_3 dz.$$

It will be exact differential and by integrating, $\phi = \phi(x, y, z)$ is obtained with one arbitrary constant.

LECTURE NO.:- 35

Problems on Gradient, Divergence, Curl and Differential Operator

Example 1. If $f(x, y, z) = 3x^2y - y^3z^2$, find $\text{grad } f$ at the point $(1, -2, 1)$.

Solution. We have given $f(x, y, z) = 3x^2y - y^3z^2$.

$$\begin{aligned}\therefore \text{grad } f &= \nabla f = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) (3x^2y - y^3z^2) \\ &= \hat{i} \frac{\partial}{\partial x} (3x^2y - y^3z^2) + \hat{j} \frac{\partial}{\partial y} (3x^2y - y^3z^2) + \hat{k} \frac{\partial}{\partial z} (3x^2y - y^3z^2) \\ &= \hat{i} (6xy) + \hat{j} (3x^2 - 3y^2z^2) - \hat{k} (2y^3z) \\ &= 6xy\hat{i} + (3x^2 - 3y^2z^2)\hat{j} - 2y^3z\hat{k} \quad \dots\dots\dots (1)\end{aligned}$$

Putting $x = 1, y = -2$ and $z = 1$ in (1), we get

$$(\text{grad } f)_{(1, -2, 1)} = 6(1)(-2)\hat{i} + \{3(1)^2 - 3(-2)^2(1)^2\}\hat{j} - 2(-2)^3(1)\hat{k}.$$

$$\therefore (\text{grad } f)_{(1, -2, 1)} = -12\hat{i} - 9\hat{j} + 16\hat{k}. \quad \underline{\text{Ans.}}$$

Example 2. If $u = x + y + z, v = x^2 + y^2 + z^2, w = yz + zx + xy$, prove that

$$(\text{grad } u) \cdot [(\text{grad } v) \times (\text{grad } w)] = 0$$

Solution. We have given $u = x + y + z, v = x^2 + y^2 + z^2, w = yz + zx + xy$.

$$\therefore \text{grad } u = \frac{\partial u}{\partial x} \hat{i} + \frac{\partial u}{\partial y} \hat{j} + \frac{\partial u}{\partial z} \hat{k} = \hat{i} + \hat{j} + \hat{k},$$

$$\text{grad } v = \frac{\partial v}{\partial x} \hat{i} + \frac{\partial v}{\partial y} \hat{j} + \frac{\partial v}{\partial z} \hat{k} = 2x\hat{i} + 2y\hat{j} + 2z\hat{k}$$

$$\& \text{grad } w = \frac{\partial w}{\partial x} \hat{i} + \frac{\partial w}{\partial y} \hat{j} + \frac{\partial w}{\partial z} \hat{k} = (z + y)\hat{i} + (z + x)\hat{j} + (y + x)\hat{k}$$

$$\begin{aligned}\therefore (\text{grad } u) \cdot [(\text{grad } v) \times (\text{grad } w)] &= \begin{vmatrix} 1 & 1 & 1 \\ 2x & 2y & 2z \\ z+y & z+x & y+x \end{vmatrix} \\ &= 2 \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ z+y & z+x & y+x \end{vmatrix} \\ &= 2 \begin{vmatrix} 1 & 1 & 1 \\ x+y+z & x+y+z & x+y+z \\ z+y & z+x & y+x \end{vmatrix}, \text{ by } R_2 + R_3\end{aligned}$$

$$= 2(x+y+z) \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ z+y & z+x & y+x \end{vmatrix}$$

$= 2(x+y+z).0$, since first two rows of determinant are same

$\therefore (\text{grad } u) \cdot [(\text{grad } v) \times (\text{grad } w)] = 0$. Proved.

Example 3. If $r = |\vec{r}|$, where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, prove that

$$(i) \quad \nabla f(r) = f'(r) \nabla r, \quad (ii) \quad \nabla r = \frac{1}{r} \vec{r}, \quad (iii) \quad \nabla f(r) \times \vec{r} = \vec{0},$$

$$(iv) \quad \nabla \left(\frac{1}{r} \right) = -\frac{\vec{r}}{r^3}, \quad (v) \quad \nabla \log |\vec{r}| = \frac{\vec{r}}{r^2}, \quad (vi) \quad \nabla r^n = nr^{n-2} \vec{r}.$$

Solution. We have given $r = |\vec{r}|$, where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ (1)

$$\begin{aligned} (i) \quad \nabla f(r) &= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) f(r) = \hat{i} \frac{\partial}{\partial x} f(r) + \hat{j} \frac{\partial}{\partial y} f(r) + \hat{k} \frac{\partial}{\partial z} f(r) \\ &= \hat{i} f'(r) \frac{\partial r}{\partial x} + \hat{j} f'(r) \frac{\partial r}{\partial y} + \hat{k} f'(r) \frac{\partial r}{\partial z} = f'(r) \left(\hat{i} \frac{\partial r}{\partial x} + \hat{j} \frac{\partial r}{\partial y} + \hat{k} \frac{\partial r}{\partial z} \right) \\ &= f'(r) \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) r = f'(r) \nabla r. \end{aligned}$$

$\therefore \nabla f(r) = f'(r) \nabla r$. Proved.

$$(ii) \quad \nabla r = \left(\hat{i} \frac{\partial r}{\partial x} + \hat{j} \frac{\partial r}{\partial y} + \hat{k} \frac{\partial r}{\partial z} \right). \quad \dots \dots \dots (2)$$

$$\text{Now from (1), } r = |x\hat{i} + y\hat{j} + z\hat{k}| = \sqrt{x^2 + y^2 + z^2} \Rightarrow r^2 = x^2 + y^2 + z^2. \quad \dots \dots \dots (3)$$

Differentiating (3) partially w.r.t. x , we get

$$2r \frac{\partial r}{\partial x} = 2x \Rightarrow \frac{\partial r}{\partial x} = \frac{x}{r}$$

$$\text{Similarly } \frac{\partial r}{\partial y} = \frac{y}{r}, \quad \frac{\partial r}{\partial z} = \frac{z}{r}.$$

Putting these values in (2), we get

$$\nabla r = \left(\hat{i} \frac{x}{r} + \hat{j} \frac{y}{r} + \hat{k} \frac{z}{r} \right) = \frac{1}{r} (x\hat{i} + y\hat{j} + z\hat{k}) = \frac{1}{r} \vec{r} = \hat{r}.$$

$\therefore \nabla r = \frac{1}{r} \vec{r}$. Proved.

(iii) From (i), $\nabla f(r) = f'(r)\nabla r$ and from (ii), $\nabla r = \frac{1}{r}\bar{r}$.

$$\therefore \nabla f(r) = f'(r)\frac{\bar{r}}{r}$$

$$\Rightarrow \nabla f(r) \times \bar{r} = \left\{ f'(r)\frac{\bar{r}}{r} \right\} \times \bar{r} = \left\{ \frac{f'(r)}{r} \right\} (\bar{r} \times \bar{r}) = \bar{0}, \text{ since } \bar{r} \times \bar{r} = \bar{0}.$$

$$\therefore \nabla f(r) \times \bar{r} = \bar{0}. \text{ Proved.}$$

$$\begin{aligned} \text{(iv)} \quad \nabla \left(\frac{1}{r} \right) &= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \left(\frac{1}{r} \right) = \left\{ \hat{i} \frac{\partial}{\partial x} \left(\frac{1}{r} \right) + \hat{j} \frac{\partial}{\partial y} \left(\frac{1}{r} \right) + \hat{k} \frac{\partial}{\partial z} \left(\frac{1}{r} \right) \right\} \\ &= \left\{ \hat{i} \left(-\frac{1}{r^2} \right) \frac{\partial r}{\partial x} + \hat{j} \left(-\frac{1}{r^2} \right) \frac{\partial r}{\partial y} + \hat{k} \left(-\frac{1}{r^2} \right) \frac{\partial r}{\partial z} \right\} \\ &= -\frac{1}{r^2} \left\{ \hat{i} \frac{x}{r} + \hat{j} \frac{y}{r} + \hat{k} \frac{z}{r} \right\}, \text{ since } \frac{\partial r}{\partial x} = \frac{x}{r}, \frac{\partial r}{\partial y} = \frac{y}{r}, \frac{\partial r}{\partial z} = \frac{z}{r} \\ &= -\frac{1}{r^3} (\hat{i}x + \hat{j}y + \hat{k}z) = -\frac{\bar{r}}{r^3}. \end{aligned}$$

$$\therefore \nabla \left(\frac{1}{r} \right) = -\frac{\bar{r}}{r^3}. \text{ Proved.}$$

$$\begin{aligned} \text{(v)} \quad \nabla \log |\bar{r}| &= \nabla \log r = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \log r \\ &= \hat{i} \frac{\partial}{\partial x} (\log r) + \hat{j} \frac{\partial}{\partial y} (\log r) + \hat{k} \frac{\partial}{\partial z} (\log r) = \hat{i} \left(\frac{1}{r} \right) \frac{\partial r}{\partial x} + \hat{j} \left(\frac{1}{r} \right) \frac{\partial r}{\partial y} + \hat{k} \left(\frac{1}{r} \right) \frac{\partial r}{\partial z} \\ &= \frac{1}{r} \left(\hat{i} \frac{x}{r} + \hat{j} \frac{y}{r} + \hat{k} \frac{z}{r} \right), \text{ since } \frac{\partial r}{\partial x} = \frac{x}{r}, \frac{\partial r}{\partial y} = \frac{y}{r}, \frac{\partial r}{\partial z} = \frac{z}{r} \\ &= \frac{1}{r^2} (x\hat{i} + y\hat{j} + z\hat{k}) = \frac{\bar{r}}{r^2}. \end{aligned}$$

$$\therefore \nabla \log |\bar{r}| = \frac{\bar{r}}{r^2}. \text{ Proved.}$$

$$\begin{aligned} \text{(vi)} \quad \nabla r^n &= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) r^n = \left(\hat{i} \frac{\partial r^n}{\partial x} + \hat{j} \frac{\partial r^n}{\partial y} + \hat{k} \frac{\partial r^n}{\partial z} \right) \\ &= nr^{n-1} \frac{\partial r}{\partial x} \hat{i} + nr^{n-1} \frac{\partial r}{\partial y} \hat{j} + nr^{n-1} \frac{\partial r}{\partial z} \hat{k} = nr^{n-1} \left(\frac{\partial r}{\partial x} \hat{i} + \frac{\partial r}{\partial y} \hat{j} + \frac{\partial r}{\partial z} \hat{k} \right) \\ &= nr^{n-1} \left(\frac{x}{r} \hat{i} + \frac{y}{r} \hat{j} + \frac{z}{r} \hat{k} \right), \text{ since } \frac{\partial r}{\partial x} = \frac{x}{r}, \frac{\partial r}{\partial y} = \frac{y}{r}, \frac{\partial r}{\partial z} = \frac{z}{r} \end{aligned}$$

$$= nr^{n-2} (x\hat{i} + y\hat{j} + z\hat{k}) = nr^{n-2}\vec{r}.$$

$$\therefore \nabla r^n = nr^{n-2}\vec{r}. \text{ Proved.}$$

Example 4. If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, then prove that (i) $\text{div } \vec{r} = 3$, (ii) $\text{curl } \vec{r} = \vec{0}$.

Solution. We have

$$\begin{aligned} \text{(i) } \text{div } \vec{r} &= \nabla \cdot \vec{r} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot (x\hat{i} + y\hat{j} + z\hat{k}) \\ &= \hat{i} \cdot \hat{i} \frac{\partial x}{\partial x} + \hat{j} \cdot \hat{j} \frac{\partial y}{\partial y} + \hat{k} \cdot \hat{k} \frac{\partial z}{\partial z} = 1 + 1 + 1 = 3. \end{aligned}$$

$$\therefore \text{div } \vec{r} = 3. \text{ Proved.}$$

$$\begin{aligned} \text{(ii) } \text{curl } \vec{r} &= \nabla \times \vec{r} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \times (x\hat{i} + y\hat{j} + z\hat{k}) \\ &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{vmatrix} = \hat{i} \left(\frac{\partial z}{\partial y} - \frac{\partial y}{\partial z} \right) + \hat{j} \left(\frac{\partial x}{\partial z} - \frac{\partial z}{\partial x} \right) + \hat{k} \left(\frac{\partial y}{\partial x} - \frac{\partial x}{\partial y} \right) = \hat{i}(0) + \hat{j}(0) + \hat{k}(0) \end{aligned}$$

$$\therefore \text{curl } \vec{r} = \vec{0}. \text{ Proved.}$$

Example 5. If $\vec{V} = \frac{x\hat{i} + y\hat{j} + z\hat{k}}{\sqrt{x^2 + y^2 + z^2}}$, then find the value of $\text{div } \vec{V}$ and $\text{curl } \vec{V}$.

Solution. We have

$$\begin{aligned} \text{div } \vec{V} &= \nabla \cdot \vec{V} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot \left\{ \frac{x\hat{i} + y\hat{j} + z\hat{k}}{\sqrt{x^2 + y^2 + z^2}} \right\} \\ &= \hat{i} \cdot \frac{\partial}{\partial x} \left(\frac{x\hat{i} + y\hat{j} + z\hat{k}}{\sqrt{x^2 + y^2 + z^2}} \right) + \hat{j} \cdot \frac{\partial}{\partial y} \left(\frac{x\hat{i} + y\hat{j} + z\hat{k}}{\sqrt{x^2 + y^2 + z^2}} \right) + \hat{k} \cdot \frac{\partial}{\partial z} \left(\frac{x\hat{i} + y\hat{j} + z\hat{k}}{\sqrt{x^2 + y^2 + z^2}} \right) \\ &= \frac{\partial}{\partial x} \left(\frac{x}{\sqrt{x^2 + y^2 + z^2}} \right) + \frac{\partial}{\partial y} \left(\frac{y}{\sqrt{x^2 + y^2 + z^2}} \right) + \frac{\partial}{\partial z} \left(\frac{z}{\sqrt{x^2 + y^2 + z^2}} \right) \\ &= \sum \frac{1 \cdot \sqrt{x^2 + y^2 + z^2} - x \cdot \frac{1}{2} (x^2 + y^2 + z^2)^{-1/2} \cdot 2x}{x^2 + y^2 + z^2} = \sum \frac{\sqrt{x^2 + y^2 + z^2} - x^2 (x^2 + y^2 + z^2)^{-1/2}}{x^2 + y^2 + z^2} \\ &= \frac{\sqrt{x^2 + y^2 + z^2} - x^2 (x^2 + y^2 + z^2)^{-1/2} + \sqrt{x^2 + y^2 + z^2} - y^2 (x^2 + y^2 + z^2)^{-1/2} + \sqrt{x^2 + y^2 + z^2} - z^2 (x^2 + y^2 + z^2)^{-1/2}}{x^2 + y^2 + z^2} \\ &= \frac{3\sqrt{x^2 + y^2 + z^2} - (x^2 + y^2 + z^2)^{-\frac{1}{2}} (x^2 + y^2 + z^2)}{x^2 + y^2 + z^2} = \frac{3\sqrt{x^2 + y^2 + z^2} - (x^2 + y^2 + z^2)^{\frac{1}{2}}}{x^2 + y^2 + z^2} = \frac{2\sqrt{x^2 + y^2 + z^2}}{x^2 + y^2 + z^2} \end{aligned}$$

$$\therefore \text{div } \vec{V} = \frac{2}{\sqrt{x^2 + y^2 + z^2}}. \text{ Ans.}$$

$$\begin{aligned}
 \text{curl } \bar{V} &= \nabla \times \bar{V} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \times \left(\frac{x\hat{i} + y\hat{j} + z\hat{k}}{\sqrt{x^2 + y^2 + z^2}} \right) \\
 &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{x}{\sqrt{x^2 + y^2 + z^2}} & \frac{y}{\sqrt{x^2 + y^2 + z^2}} & \frac{z}{\sqrt{x^2 + y^2 + z^2}} \end{vmatrix} \\
 &= \sum \hat{i} \left[\frac{\partial}{\partial y} \left(\frac{z}{\sqrt{x^2 + y^2 + z^2}} \right) - \frac{\partial}{\partial z} \left(\frac{y}{\sqrt{x^2 + y^2 + z^2}} \right) \right] \\
 &= \sum \hat{i} \left[z \left\{ -\frac{1}{2} \frac{2y}{(x^2 + y^2 + z^2)^{3/2}} \right\} - y \left\{ -\frac{1}{2} \frac{2z}{(x^2 + y^2 + z^2)^{3/2}} \right\} \right] \\
 &= \sum \hat{i} \left[-\frac{yz}{(x^2 + y^2 + z^2)^{3/2}} + \frac{yz}{(x^2 + y^2 + z^2)^{3/2}} \right] = \sum \hat{i} 0 = \hat{i}0 + \hat{j}0 + \hat{k}0 = \bar{0}.
 \end{aligned}$$

$\therefore \text{curl } \bar{V} = \bar{0}$. Ans.

Example 6. Find the value of n for which vector $r^n \bar{r}$ is solenoidal, where $\bar{r} = x\hat{i} + y\hat{j} + z\hat{k}$.

Solution. A vector $\bar{V}$ is said to be solenoidal, if $\text{div } \bar{V} = 0$.

Here $\bar{V} = r^n \bar{r}$, where $\bar{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $r = |\bar{r}| = |x\hat{i} + y\hat{j} + z\hat{k}| = \sqrt{x^2 + y^2 + z^2}$.

$$\begin{aligned}
 \therefore \nabla \cdot (r^n \bar{r}) &= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot \left[(x^2 + y^2 + z^2)^{\frac{n}{2}} (x\hat{i} + y\hat{j} + z\hat{k}) \right] \\
 &= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot \left[x(x^2 + y^2 + z^2)^{\frac{n}{2}} \hat{i} + y(x^2 + y^2 + z^2)^{\frac{n}{2}} \hat{j} + z(x^2 + y^2 + z^2)^{\frac{n}{2}} \hat{k} \right] \\
 &= \sum \frac{\partial}{\partial x} \left[x(x^2 + y^2 + z^2)^{\frac{n}{2}} \right] \\
 &= \sum \left[1(x^2 + y^2 + z^2)^{\frac{n}{2}} + x \frac{n}{2} (x^2 + y^2 + z^2)^{\frac{n}{2}-1} 2x \right] \\
 &= \sum \left[(x^2 + y^2 + z^2)^{\frac{n}{2}} + nx^2 (x^2 + y^2 + z^2)^{\frac{n}{2}-1} \right] \\
 &= (x^2 + y^2 + z^2)^{\frac{n}{2}} \sum 1 + n(x^2 + y^2 + z^2)^{\frac{n}{2}-1} \sum x^2 \\
 &= (x^2 + y^2 + z^2)^{\frac{n}{2}} (1 + 1 + 1) + n(x^2 + y^2 + z^2)^{\frac{n}{2}-1} (x^2 + y^2 + z^2) \\
 &= 3(x^2 + y^2 + z^2)^{\frac{n}{2}} + n(x^2 + y^2 + z^2)^{\frac{n}{2}} \\
 &= (n + 3)(x^2 + y^2 + z^2)^{\frac{n}{2}}
 \end{aligned}$$

If the vector $r^n \bar{r}$ is solenoidal, then $\nabla \cdot (r^n \bar{r}) = 0$ i.e., $(n + 3)(x^2 + y^2 + z^2)^{\frac{n}{2}} = 0$ or $n + 3 = 0 \Rightarrow n = -3$. Ans.

Example 7. Determine the constant a so that the vector $\bar{V} = (x+3y)\hat{i} + (y-2z)\hat{j} + (x+az)\hat{k}$ is solenoidal.

Solution. A vector $\bar{V}$ is said to be solenoidal, if $\text{div } \bar{V} = 0$. We have

$$\begin{aligned}\text{div } \bar{V} = \nabla \cdot \bar{V} &= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot \{ (x+3y)\hat{i} + (y-2z)\hat{j} + (x+az)\hat{k} \} \\ &= \frac{\partial}{\partial x}(x+3y) + \frac{\partial}{\partial y}(y-2z) + \frac{\partial}{\partial z}(x+az) \\ &= 1+1+a = 2+a\end{aligned}$$

Now $\text{div } \bar{V} = 0$, if $2+a=0 \Rightarrow a=-2$. Ans.

Example 8. Prove that $\text{div} \left(\frac{\bar{r}}{r^3} \right) = 0$.

Solution. We have

$$\begin{aligned}\text{div} \left(\frac{\bar{r}}{r^3} \right) &= \text{div} (r^{-3} \bar{r}) \\ &= (\text{grad } r^{-3}) \cdot \bar{r} + r^{-3} \text{div } \bar{r}, \text{ by } \text{div}(\phi \bar{A}) = (\text{grad } \phi) \cdot \bar{A} + \phi \text{div } \bar{A} \\ &= (-3r^{-5} \bar{r}) \cdot \bar{r} + r^{-3} 3, \text{ by } \nabla r^n = nr^{n-2} \bar{r} \text{ [Ex. 3(vi)] \& } \text{div } \bar{r} = 3 \text{ [Ex. 4(i)]} \\ &= 3r^{-3} - 3r^{-5}(\bar{r} \cdot \bar{r}) \\ &= 3r^{-3} - 3r^{-5}r^2 = 3r^{-3} - 3r^{-3} = 0\end{aligned}$$

$$\therefore \text{div} \left(\frac{\bar{r}}{r^3} \right) = 0. \text{ Proved.}$$

Note. We can evaluate $\text{div} \left(\frac{\bar{r}}{r^3} \right)$ by using $\bar{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and Σ form as example 6 and without using above identity. Here we observe that the vector $r^{-3}\bar{r}$ is solenoidal.

Example 9. Prove that $\text{div}(\hat{r}) = \frac{2}{r}$ where $\hat{r} = \frac{\bar{r}}{r}$ and $\bar{r} = x\hat{i} + y\hat{j} + z\hat{k}$.

Solution. Do yourself as example 8.

Example 10. A vector $\bar{r}$ is defined by $\bar{r} = x\hat{i} + y\hat{j} + z\hat{k}$. If $|\bar{r}| = r$, then show that $r^n \bar{r}$ is irrotational.

Hint. For irrotational, $\text{curl } r^n \bar{r} = \bar{0}$. Do yourself as example 5.

Example 11. If $\vec{u} = (x+2y+az)\hat{i} + (bx-3y-z)\hat{j} + (4x+cy+2z)\hat{k}$, find a, b, c so that $\vec{u}$ is irrotational. Also find its scalar potential.

Hint. For irrotational, $\text{curl } \vec{u} = \vec{0}$. Then $\vec{u} = \nabla \phi$. Do yourself.

Example 12. A fluid motion is given by $\vec{q} = (y+z)\hat{i} + (z+x)\hat{j} + (x+y)\hat{k}$. Is this motion irrotational? If so, find its velocity potential.

Solution. Try yourself.

LECTURE NO.:- 36

Scalar and Vector Line Integrals

Let $P(x, y, z)$ be any point on a curve C , $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ be the position vector of the point P , $\phi = \phi(x, y, z)$ be a continuous scalar point function and $\vec{F} = \vec{F}(x, y, z) = F_1\hat{i} + F_2\hat{j} + F_3\hat{k}$ be a continuous vector point function, then

$$\int_C \phi \, d\vec{r} = \int_C \phi(x, y, z) (dx\hat{i} + dy\hat{j} + dz\hat{k})$$

is called the line integral of the scalar point function ϕ along the curve C and

$$\int_C \vec{F} \cdot d\vec{r} = \int_C (F_1 dx + F_2 dy + F_3 dz)$$

is called the line integral of the vector point function $\vec{F}$ along the curve C .

If $\vec{F}$ be the force acting on a particle moving on the curve C , then the line integral $\int_C \vec{F} \cdot d\vec{r}$ gives the work done by the force $\vec{F}$ in displacing the particle along the curve C .

If the work done (say W) in moving a particle from point A to point B depends only on the initial and terminal points A and B and not on the path followed to reaching B from A , then the field of the force is called a conservative field. In such a case $\text{curl } \vec{F} = \vec{0}$ and $\vec{F} = \text{grad } \phi$, where $\phi = \phi(x, y, z)$ be a differentiable scalar point function and is called a scalar potential function of $\vec{F}$. Then we have

$$\begin{aligned} W &= \int_C \vec{F} \cdot d\vec{r} = \int_C \nabla \phi \cdot d\vec{r} \\ &= \int_A^B \left(\hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y} + \hat{k} \frac{\partial \phi}{\partial z} \right) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k}) \\ &= \int_A^B \left(\frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy + \frac{\partial \phi}{\partial z} dz \right) \\ &= \int_A^B d\phi = [\phi(x, y, z)]_A^B \\ &= \phi_B - \phi_A. \end{aligned}$$

If $B = A$ (for closed curve C), then work done (W) = 0.

The line integral $\int_C \vec{V} \cdot d\vec{r}$ of the velocity $\vec{V}$ along path C is called circulation.

Another line integral is $\int_C \vec{F} \times d\vec{r}$, which may be torque or moment, if $\vec{F}$ is force.

Example 1. A vector field is given by $\vec{F} = (\sin y)\hat{i} + x(1 + \cos y)\hat{j}$. Evaluate $\int_C \vec{F} \cdot d\vec{r}$ over a circular path given by $x^2 + y^2 = a^2, z = 0$.

Solution. For given circle $x^2 + y^2 = a^2, z = 0$; we can take parametric equations as

$$x = a \cos \theta, \quad y = a \sin \theta, \quad z = 0; \quad 0 \leq \theta \leq 2\pi.$$

$$\Rightarrow dx = -a \sin \theta \, d\theta, \quad dy = a \cos \theta \, d\theta, \quad dz = 0; \quad 0 \leq \theta \leq 2\pi.$$

$$\begin{aligned}
\therefore \int_C \vec{F} \cdot d\vec{r} &= \int_C [(\sin y)\hat{i} + x(1 + \cos y)\hat{j}] \cdot [(dx)\hat{i} + (dy)\hat{j} + (dz)\hat{k}] \\
&= \int_0^{2\pi} [\{\sin(a \sin \theta)\}(-a \sin \theta) + a \cos \theta \{1 + \cos(a \sin \theta)\}(a \cos \theta)] d\theta \\
&= a \int_0^{2\pi} [\{\sin(a \sin \theta)\}d(\cos \theta) + (\cos \theta)d\{\sin(a \sin \theta)\}] + a^2 \int_0^{2\pi} \cos^2 \theta d\theta \\
&= a \int_0^{2\pi} d[(\cos \theta)\{\sin(a \sin \theta)\}] + a^2 \int_0^{2\pi} \left[\frac{1+\cos 2\theta}{2}\right] d\theta \\
&= a[(\cos \theta)\{\sin(a \sin \theta)\}]_0^{2\pi} + \frac{a^2}{2} \left[\theta + \frac{\sin 2\theta}{2}\right]_0^{2\pi} \\
&= a[(\cos 2\pi)\{\sin(a \sin 2\pi)\} - (\cos 0)\{\sin(a \sin 0)\}] + \frac{a^2}{2} \left[2\pi + \frac{0}{2} - \left\{0 + \frac{0}{2}\right\}\right] \\
&= a[(1)\{0\} - (1)\{0\}] + \frac{a^2}{2} [2\pi] = 0 + a^2\pi
\end{aligned}$$

$$\therefore \int_C \vec{F} \cdot d\vec{r} = a^2\pi. \quad \underline{\text{Ans.}}$$

Example 2. Find the constant λ so that $\vec{F}$ is a conservative vector field, where $\vec{F} = (\lambda xy - z^3)\hat{i} + (\lambda - 2)x^2\hat{j} + (1 - \lambda)xz^2\hat{k}$. Find the work done in moving a particle from $(1, 2, -3)$ to $(1, -4, 2)$ in the field.

Solution. Try yourself.

Scalar and Vector Surface Integrals

Let $P(x, y, z)$ be any point on a surface S given by $f(x, y, z) = 0$, $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ be the position vector of the point P , dS be an elementary surface of the surface S containing point P , $\hat{n}$ be outward drawn unit normal vector to the surface S at point P so that $\hat{n} = \frac{\nabla f}{|\nabla f|}$, $\vec{dS}$ be an elementary surface vector of the surface S such that $\vec{dS} = \hat{n} dS$, $\phi = \phi(x, y, z)$ be a continuous scalar point function and $\vec{F} = \vec{F}(x, y, z) = F_1\hat{i} + F_2\hat{j} + F_3\hat{k}$ be a continuous vector point function, then

$$\int_S \phi \vec{dS} = \int_S \phi(x, y, z) \hat{n} dS$$

is called the normal surface integral of the scalar point function ϕ over the surface S and

$$\int_S \vec{F} \cdot \vec{dS} = \int_S \vec{F} \cdot \hat{n} dS$$

is called the normal surface integral of the vector point function $\vec{F}$ over the surface S and is called flux of $\vec{F}$ across the surface S .

If $\hat{n}$ makes angles α , β and γ with the coordinate axes x , y and z respectively, then $\hat{n} = \hat{i} \cos \alpha + \hat{j} \cos \beta + \hat{k} \cos \gamma$ and the projection of dS on xy -plane is

$$dxdy = dS \cos \gamma = dS \hat{n} \cdot \hat{k}.$$

$$\text{So } dS = \frac{dxdy}{\hat{n} \cdot \hat{k}}.$$

Similarly $dS = \frac{dydz}{\hat{n} \cdot \hat{i}} = \frac{dzdx}{\hat{n} \cdot \hat{j}}$. Thus we have

$$\begin{aligned}\int_S \vec{F} \cdot d\vec{S} &= \int_S \vec{F} \cdot \hat{n} dS \\ &= \iint_{R_x} \vec{F} \cdot \hat{n} \frac{dydz}{\hat{n} \cdot \hat{i}} \\ &= \iint_{R_y} \vec{F} \cdot \hat{n} \frac{dzdx}{\hat{n} \cdot \hat{j}} \\ &= \iint_{R_z} \vec{F} \cdot \hat{n} \frac{dxdy}{\hat{n} \cdot \hat{k}},\end{aligned}$$

where R_x , R_y and R_z be the projections of the surface S on the coordinate planes $x = 0$, $y = 0$ and $z = 0$ respectively.

Example 3. Calculate the flux of the water through the parabolic cylinder $y = x^2$ between the planes $x = 0$, $z = 0$, $x = 3$, $z = 2$, if the velocity vector is $\vec{A} = y\hat{i} + 2\hat{j} + xz\hat{k}$ m/sec.

Solution. Given surface (parabolic cylinder) is $S: f(x, y, z) = x^2 - y = 0$.

Unit normal vector to the surface S is $\hat{n} = \frac{\nabla f}{|\nabla f|} = \frac{2x\hat{i} - \hat{j} + 0\hat{k}}{|2x\hat{i} - \hat{j}|} = \frac{2x\hat{i} - \hat{j}}{\sqrt{4x^2 + 1}}$.

Given velocity vector is $\vec{A} = y\hat{i} + 2\hat{j} + xz\hat{k}$ m/sec.

$\therefore$ Required flux of the water $= \int_S \vec{A} \cdot \hat{n} dS$

$$\begin{aligned}&= \iint_R \vec{A} \cdot \hat{n} \frac{dxdz}{|\hat{n} \cdot \hat{j}|} \\ &= \int_0^3 \int_0^2 (y\hat{i} + 2\hat{j} + xz\hat{k}) \cdot \frac{2x\hat{i} - \hat{j}}{\sqrt{4x^2 + 1}} \frac{dxdz}{\left| \frac{2x\hat{i} - \hat{j}}{\sqrt{4x^2 + 1}} \cdot \hat{j} \right|} \\ &= \int_0^3 \int_0^2 (y\hat{i} + 2\hat{j} + xz\hat{k}) \cdot (2x\hat{i} - \hat{j}) \frac{dxdz}{|-1|} \\ &= \int_0^3 \int_0^2 (y2x - 2) dxdz \\ &= \int_0^3 \int_0^2 (x^2 2x - 2) dxdz \\ &= \int_0^2 \left(\frac{2x^4}{4} - 2x \right) \Big|_0^3 dz \\ &= \int_0^2 \left(\frac{3^4}{2} - 6 - 0 \right) dz \\ &= \left(\frac{81}{2} - 6 \right) \int_0^2 dz \\ &= \frac{69}{2} (z) \Big|_0^2 \\ &= \frac{69}{2} (2 - 0) \\ &= 69 \text{ m}^3/\text{sec.} \quad \underline{\text{Ans.}}\end{aligned}$$

Example 4. Find the surface area of the plane $x + 2y + 2z = 12$ cut off by planes $x = 0, y = 0, x = 1, y = 1$.

Solution. Given surface (plane) is $S: f(x, y, z) = x + 2y + 2z - 12 = 0$.

Unit normal vector to the surface S is $\hat{n} = \frac{\nabla f}{|\nabla f|} = \frac{1\hat{i} + 2\hat{j} + 2\hat{k}}{|1\hat{i} + 2\hat{j} + 2\hat{k}|} = \frac{1\hat{i} + 2\hat{j} + 2\hat{k}}{\sqrt{1+4+4}} = \frac{1\hat{i} + 2\hat{j} + 2\hat{k}}{3}$.

$\therefore$ Required surface area $= \int_S dS$

$$\begin{aligned} &= \iint_R \frac{dxdy}{|\hat{n} \cdot \hat{k}|} \\ &= \int_0^1 \int_0^1 \frac{dxdy}{\left| \frac{1\hat{i} + 2\hat{j} + 2\hat{k}}{3} \cdot \hat{k} \right|} \\ &= \int_0^1 \int_0^1 \frac{dxdy}{2/3} \\ &= \frac{3}{2} \int_0^1 \int_0^1 dxdy \\ &= \frac{3}{2} (x)_0^1 (y)_0^1 \\ &= \frac{3}{2} \cdot \underline{\text{Ans.}} \end{aligned}$$

Scalar and Vector Volume Integrals

Suppose V is the volume bounded by a surface S . Divide the volume V into sub-volumes $\delta V_1, \delta V_2, \dots, \delta V_n$. In each δV_i , choose an arbitrary point P_i whose coordinates are (x_i, y_i, z_i) . Let φ be a single valued function defined over V . Form the sum $\sum \varphi(P_i) \delta V_i$, where $\varphi(P_i) = \varphi(x_i, y_i, z_i)$.

Now let us take the limit of the sum as $n \rightarrow \infty$, then the limit, if exists, is called the volume integral of φ over V and is denoted as $\iiint_V \varphi dV$.

Likewise if $\vec{F}$ is a vector point function defined in the given region of volume V then vector volume integral of $\vec{F}$ over V is $\iiint_V \vec{F} dV$.

Note: Above volume integral becomes $\iiint_V \varphi dx dy dz$ if we subdivide the volume V into small cuboids by drawing lines parallel to three co-ordinate axes because in that case $dV = dxdydz$.

Example 5.

If $\vec{F} = (2x^2 - 3z)\hat{i} - 2xy\hat{j} - 4x\hat{k}$, evaluate $\iiint_V \nabla \times \vec{F} dV$ where V is the region bounded by the co-ordinate planes and the plane $2x + 2y + z = 4$.

Solution: Consider

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x^2 - 3z & -2xy & -4x \end{vmatrix} = 0\hat{i} + \hat{j} - 2y\hat{k}$$

Region bounded by $2x + 2y + z = 4$ and coordinate planes such that

$$2x \leq 4, \quad 2x + 2y \leq 4, \quad 2x + 2y + z \leq 4$$

$$\text{i.e.} \quad x \leq 2, \quad y \leq 2 - x, \quad z \leq 4 - 2x - 2y$$

$$\begin{aligned} \therefore \iiint_V \nabla \times \vec{F} \, dV &= \int_{x=0}^2 \int_{y=0}^{2-x} \int_{z=0}^{4-2x-2y} (\hat{j} - 2y \hat{k}) \, dz \, dy \, dx \\ &= \int_{x=0}^2 \int_{y=0}^{2-x} [z \hat{j} - 2yz \hat{k}]_{z=0}^{z=4-2x-2y} \, dy \, dx \\ &= \int_{x=0}^2 \int_{y=0}^{2-x} \{(4 - 2x - 2y) \hat{j} - 2y(4 - 2x - 2y) \hat{k}\} \, dy \, dx \\ &= \int_{x=0}^2 \left[(4y - 2xy - y^2) \hat{j} - \left(4y^2 - 2xy^2 - \frac{4}{3}y^3 \right) \hat{k} \right]_{y=0}^{y=2-x} \, dx \\ &= \int_{x=0}^2 \left[\{8 - 4x - 2x(2 - x) - (2 - x)^2\} \hat{j} - \left(4(2 - x)^2 - 2x(2 - x)^2 - \frac{4}{3}(2 - x)^3 \right) \hat{k} \right] \, dx \\ &= \int_{x=0}^2 \left[(4 - 4x + x^2) \hat{j} - \left(-\frac{2}{3}x^3 + 4x^2 - 8x + \frac{16}{3} \right) \hat{k} \right] \, dx \\ &= \left[4x - 2x^2 + \frac{x^3}{3} \right]_{x=0}^{x=2} \hat{j} - \left[-\frac{1}{6}x^4 + \frac{4}{3}x^3 - 4x^2 + \frac{16}{3}x \right]_{x=0}^{x=2} \hat{k} \\ &= \frac{8}{3} \hat{j} - \frac{8}{3} \hat{k} = \frac{8}{3} (\hat{j} - \hat{k}) \end{aligned}$$

Example 6. Integrate $\nabla \cdot \vec{F}$ over the unit cube $0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq 1$, where $\vec{F}$ is the vector function $x^2y \hat{i} + (x - z) \hat{j} + 2xz^2 \hat{k}$.

Solution. Given $\vec{F} = x^2y \hat{i} + (x - z) \hat{j} + 2xz^2 \hat{k}$. So we have

$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(x^2y) + \frac{\partial}{\partial y}(x - z) + \frac{\partial}{\partial z}(2xz^2) = 2xy + 4xz.$$

The required integral is

$$\begin{aligned} &\int_{x=0}^1 \int_{y=0}^1 \int_{z=0}^1 (2xy + 4xz) \, dz \, dy \, dx = \int_{x=0}^1 \int_{y=0}^1 [2xyz + 2xz^2]_0^1 \, dy \, dx \\ &= \int_{x=0}^1 \int_{y=0}^1 [2xy + 2x] \, dy \, dx = \int_{x=0}^1 [xy^2 + 2xy]_0^1 \, dx \\ &= \int_{x=0}^1 3x \, dx = \left[\frac{3}{2}x^2 \right]_0^1 = \frac{3}{2} \end{aligned}$$

LECTURE NO.:- 37

Problems of Line, Surface and Volume Integrals

Example 7.

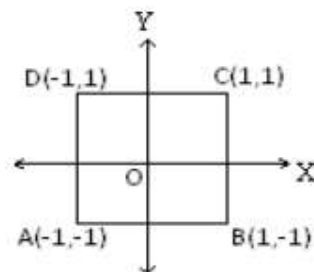
Evaluate the line integral $\int [(x^2 + xy)dx + (x^2 + y^2)dy]$, where C is the square whose vertices are $(-1, -1)$, $(1, -1)$, $(1, 1)$ and $(-1, 1)$.

Solution: Curve C is square in the xy plane where $z = 0$

$\therefore \vec{r} = x\hat{i} + y\hat{j}$ in xy plane

$$d\vec{r} = dx\hat{i} + dy\hat{j} \quad \dots (1)$$

$$\text{Now } \int_C \vec{F} \cdot d\vec{r} = \int [(x^2 + xy)dx + (x^2 + y^2)dy]$$



Path of the integration is shown in figure, it consists of lines AB, BC, CD and DA. As curve C is a square, then

On AB, $y = -1 \Rightarrow dy = 0$ and x varies from -1 to 1

On BC, $x = 1 \Rightarrow dx = 0$ and y varies from -1 to 1

On CD, $y = 1 \Rightarrow dy = 0$ and x varies from 1 to -1

On DA, $x = -1 \Rightarrow dx = 0$ and y varies from 1 to -1

$$\begin{aligned} \int_C \vec{F} \cdot d\vec{r} &= \int_{AB} (x^2 - x)dx + \int_{BC} (1 + y^2)dy + \int_{CD} (x^2 + x)dx + \int_{DA} (1 + y^2)dy \\ &= \int_{-1}^1 (x^2 - x)dx + \int_{-1}^1 (1 + y^2)dy + \int_1^{-1} (x^2 + x)dx + \int_1^{-1} (1 + y^2)dy \\ &= \left[\frac{x^3}{3} - \frac{x^2}{2} \right]_{-1}^1 + \int_{-1}^1 (1 + y^2)dy + \left[\frac{x^3}{3} + \frac{x^2}{2} \right]_1^{-1} - \int_{-1}^1 (1 + y^2)dy \\ &= \left(\frac{1}{3} - \frac{1}{2} + \frac{1}{3} + \frac{1}{2} \right) + \left(-\frac{1}{3} + \frac{1}{2} - \frac{1}{3} - \frac{1}{2} \right) = 0 \end{aligned}$$

Example 8.

If $\vec{F} = 3xy\hat{i} - y^2\hat{j}$, evaluate $\int_C \vec{F} \cdot d\vec{r}$ where C is the arc of the parabola $y = 2x^2$ from $(0, 0)$ to $(1, 2)$.

Solution: Because the integration is performed in the xy -plane ($z = 0$), we take

$\vec{r} = x\hat{i} + y\hat{j}$ so that $d\vec{r} = dx\hat{i} + dy\hat{j}$

$$\therefore \vec{F} \cdot d\vec{r} = (3xy \hat{i} - y^2 \hat{j}) \cdot (dx \hat{i} + dy \hat{j}) = 3xy dx - y^2 dy$$

On the curve C: $y = 2x^2$ from $(0, 0)$ to $(1, 2)$

$$\vec{F} \cdot d\vec{r} = 3x(2x^2)dx - (2x^2)^2 dx = (6x^3 - 4x^4)dx$$

Also x varies from 0 to 1.

$$\int_C \vec{F} \cdot d\vec{r} = \int_0^1 (6x^3 - 4x^4)dx = \left[\frac{6x^4}{4} - \frac{4x^5}{5} \right]_0^1 = \frac{3}{2} - \frac{4}{5} = \frac{7}{10}$$

Note: If the curve is traversed in the opposite direction, that is from $(1, 2)$ to $(0, 0)$, the value of the integral would be $-\frac{7}{10}$.

Example 9.

Compute the line integral $\int_C (y^2 dx - x^2 dy)$ about the triangle whose vertices are $(1, 0)$, $(0, 1)$ and $(-1, 0)$.

Solution: Here the closed curve C is a triangle ABC.

On AB: Equation of line AB is

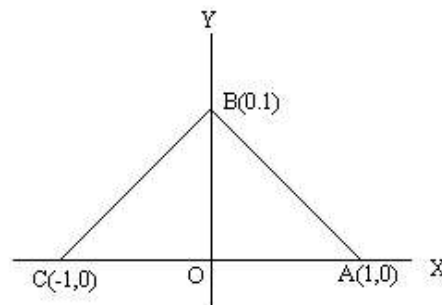
$$y - 0 = \frac{1-0}{0-1}(x - 1) \Rightarrow y = 1 - x$$

$$\therefore dy = -dx \text{ and } x \text{ varies from } 1 \text{ to } 0.$$

On BC: Equation of line BC is

$$y - 1 = \frac{0-1}{-1-0}(x - 0) \Rightarrow y = 1 + x$$

$$\therefore dy = dx \text{ and } x \text{ varies from } 0 \text{ to } -1.$$



On CA: $y = 0$. Therefore, $dy = 0$ and x varies from -1 to 1 .

$$\begin{aligned} \int_C (y^2 dx - x^2 dy) &= \int_{AB} (y^2 dx - x^2 dy) + \int_{BC} (y^2 dx - x^2 dy) + \int_{CA} (y^2 dx - x^2 dy) \\ &= \int_1^0 [(1-x)^2 dx - x^2(-dx)] + \int_0^{-1} [(1+x)^2 dx - x^2 dx] + \int_{-1}^1 0 dx \\ &= -2/3. \text{ Ans.} \end{aligned}$$

Example 10.

Find the circulation of $\vec{F}$ around the curve C where $\vec{F} = y\hat{i} + z\hat{j} + x\hat{k}$ and C is the circle $x^2 + y^2 = 1$, $z = 0$.

Solution: Circulation of $\vec{F}$ along the curve C is $\oint_C \vec{F} \cdot d\vec{r}$

Equation of circle is $x^2 + y^2 = 1, z = 0$

Its parametric equations are $x = \cos \theta, y = \sin \theta, z = 0$

Now $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k} = \cos \theta \hat{i} + \sin \theta \hat{j} + 0\hat{k}$ so that

$$d\vec{r} = (-\sin \theta \hat{i} + \cos \theta \hat{j} + 0\hat{k})d\theta$$

Also $\vec{F} = y\hat{i} + z\hat{j} + x\hat{k} = \sin \theta \hat{i} + 0\hat{j} + \cos \theta \hat{k}$

$$\begin{aligned} \therefore \oint_C \vec{F} \cdot d\vec{r} &= \oint_C (\sin \theta \hat{i} + 0\hat{j} + \cos \theta \hat{k}) \cdot (-\sin \theta \hat{i} + \cos \theta \hat{j} + 0\hat{k})d\theta \\ &= \int_0^{2\pi} -\sin^2 \theta d\theta \quad (\text{since along the circle, } \theta \text{ varies from } 0 \text{ to } 2\pi) \\ &= -\int_0^{2\pi} \frac{1 - \cos 2\theta}{2} d\theta = -\left[\frac{\theta}{2} - \frac{\sin 2\theta}{4}\right]_0^{2\pi} \\ &= -\left[\left(\frac{2\pi}{2} - 0\right) - (0 - 0)\right] = -\pi \end{aligned}$$

Example 11.

Find the work done in moving a particle once round a circle C in the xy plane, if the circle has its centre at the origin and radius 2 units and the force field is given as

$$\vec{F} = (2x - y + 2z)\hat{i} + (x + y - z)\hat{j} + (3x - 2y - 5z)\hat{k}.$$

Solution: Equation of a circle having centre $(0,0)$ with radius 2 in xy plane is $x^2 + y^2 = 4$. Parametric equations of this circle are $x = 2 \cos t, y = 2 \sin t, z = 0$.

Since integration is to be performed around a circle in xy plane,

$$\therefore \vec{r} = x\hat{i} + y\hat{j} = 2 \cos t \hat{i} + 2 \sin t \hat{j} \quad \Rightarrow \quad \frac{d\vec{r}}{dt} = -2 \sin t \hat{i} + 2 \cos t \hat{j}$$

Work done, $\int_C \left(\vec{F} \cdot \frac{d\vec{r}}{dt}\right) dt$

$$\begin{aligned} &= \int \{(2x - y + 2z)\hat{i} + (x + y - z)\hat{j} + (3x - 2y - 5z)\hat{k}\} \cdot \{-2 \sin t \hat{i} + 2 \cos t \hat{j}\} dt \\ &= \int \{(4 \cos t - 2 \sin t)\hat{i} + (2 \cos t + 2 \sin t)\hat{j} + (6 \cos t - 4 \sin t)\hat{k}\} \cdot \{-2 \sin t \hat{i} + 2 \cos t \hat{j}\} dt \end{aligned}$$

In moving round the circle, t varies from 0 to 2π

$$\begin{aligned}
 \therefore \text{Work done} &= \int_0^{2\pi} \{ (4 \cos t - 2 \sin t)(-2 \sin t) + (2 \cos t + 2 \sin t)(2 \cos t) \} dt \\
 &= \int_0^{2\pi} [-8 \cos t \sin t + 4 \sin^2 t + 4 \cos^2 t + 4 \sin t \cos t] dt \\
 &= \int_0^{2\pi} [4 - 4 \sin t \cos t] dt = \left[4t - 4 \frac{\sin^2 t}{2} \right]_0^{2\pi} \\
 &= [(8\pi - 2 \sin^2 2\pi) - (0 - 0)] = 8\pi
 \end{aligned}$$

Example 12.

Evaluate $\iint_S \vec{F} \cdot \hat{n} dS$ where $\vec{F} = z\hat{i} + x\hat{j} + 3y^2z\hat{k}$ and S is the surface of the cylinder $x^2 + y^2 = 16$ included in the first octant between $z = 0$ and $z = 5$.

Solution: A vector normal to the surface S is given by

$$\vec{n} = \nabla(x^2 + y^2) = 2x\hat{i} + 2y\hat{j} \quad \dots (1)$$

$$\therefore \hat{n} = \text{unit vector normal to surface } S \text{ at any point } (x, y, z) = \frac{2x\hat{i} + 2y\hat{j}}{\sqrt{4x^2 + 4y^2}} \quad \dots (2)$$

$$\therefore x^2 + y^2 = 16, \text{ therefore } \hat{n} = \frac{2x\hat{i} + 2y\hat{j}}{\sqrt{4(x^2 + y^2)}} = \frac{2x\hat{i} + 2y\hat{j}}{8} = \frac{x}{4}\hat{i} + \frac{y}{4}\hat{j} \quad \dots (3)$$

$$\text{Now } \iint_S \vec{F} \cdot \hat{n} dS = \iint_R \vec{F} \cdot \hat{n} \frac{dx dz}{|\hat{n} \cdot \hat{j}|}$$

(projection on xy plane can't be taken as the surface s is perpendicular to xy plane)

$$= \iint_R \left(\frac{xz}{4} + \frac{xy}{4} \right) \frac{dx dz}{\frac{y}{4}} = \iint_R \left(\frac{xz}{y} + x \right) dx dz, \quad \left(\text{since from (3), } \hat{n} \cdot \hat{j} = \frac{y}{4} \right)$$

$$= \int_{z=0}^5 \int_{x=0}^4 \left(\frac{xz}{\sqrt{16-x^2}} + x \right) dx dz = \int_{z=0}^5 \int_{x=0}^4 \left(\frac{-z}{2} \frac{(-2x)}{\sqrt{16-x^2}} + x \right) dx dz$$

$$= \int_{z=0}^5 \left[-\frac{z}{2} \frac{\sqrt{16-x^2}}{1/2} + \frac{x^2}{2} \right]_{x=0}^4 dz = \int_{z=0}^5 (4z + 8) dz = \left[\frac{4z^2}{2} + 8z \right]_{z=0}^5 = 90$$

Example 13. Evaluate $\iint_S \phi \hat{n} dS$ where $\phi = \frac{3}{8}xyz$ and S is the surface of cylinder $x^2 + y^2 = 16$ included in the first octant between $z = 0$ to $z = 5$.

Solution: A vector normal to the surface S is given by

$$\vec{n} = \nabla(x^2 + y^2) = 2x\hat{i} + 2y\hat{j}$$

$\therefore \hat{n} = \text{unit vector normal to surface } S \text{ at any point } (x, y, z) = \frac{2x\hat{i} + 2y\hat{j}}{\sqrt{4x^2 + 4y^2}}$

$$\hat{n} = \frac{2x\hat{i} + 2y\hat{j}}{\sqrt{4(x^2 + y^2)}} = \frac{2x\hat{i} + 2y\hat{j}}{8} = \frac{x}{4}\hat{i} + \frac{y}{4}\hat{j} \quad [\because x^2 + y^2 = 16]$$

$$\text{Now } \iint_S \phi \hat{n} dS = \iint_R \frac{3}{8}xyz \left(\frac{x}{4}\hat{i} + \frac{y}{4}\hat{j} \right) \frac{dx dz}{|\hat{n} \cdot \hat{j}|} \quad \dots(1)$$

where R is the region of projection of S on zx plane. Therefore, from (1)

$$\begin{aligned} \iint_S \phi \hat{n} dS &= \int_{z=0}^5 \int_{x=0}^4 \left(\frac{3}{32}x^2yz \hat{i} + \frac{3}{32}xy^2z \hat{j} \right) \frac{dx dz}{y/4}, \quad \text{where } y^2 = 16 - x^2 \\ &= \int_{z=0}^5 \int_{x=0}^4 \left(\frac{3}{8}x^2z \hat{i} + \frac{3}{8}xz\sqrt{(16 - x^2)} \hat{j} \right) dx dz \\ &= \frac{3}{8} \int_{z=0}^5 \left(\frac{x^3z}{3} \hat{i} - \frac{z(16 - x^2)^{3/2}}{3/2} \hat{j} \right)_{x=0}^4 dz \\ &= \frac{3}{8} \int_{z=0}^5 \left(\frac{64}{3}z \hat{i} + \frac{64}{3}z \hat{j} \right) dz = 8 \left[\frac{z^2}{2} \hat{i} + \frac{z^2}{2} \hat{j} \right]_{z=0}^5 = 100 \hat{i} + 100 \hat{j} \end{aligned}$$

Example 14. If $\vec{F} = 2xz \hat{i} - x \hat{j} + y^2 \hat{k}$, evaluate $\iiint_V \vec{F} dV$, where V is the region bounded by $x = 0, y = 0, x = 2, y = 6, z = x^2, z = 4$.

Solution. Try yourself.

LECTURE NO.:- 38

Green's Theorem in a Plane

If $M(x, y)$ and $N(x, y)$ are continuously differentiable functions in a region S in xy -plane bounded by a simple closed curve C , then

$$\int_C (Mdx + Ndy) = \iint_S \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dxdy,$$

where C is taken in positive sense (anticlockwise direction).

Stoke's Theorem

If $\vec{F}$ is a continuously differentiable vector point function in a region containing surface S bounded by a simple closed curve C , then

$$\int_C \vec{F} \cdot d\vec{r} = \int_S (\nabla \times \vec{F}) \cdot \hat{n} dS,$$

where $\hat{n}$ is the outward drawn unit normal vector to the surface S and drawn in the sense in which a right handed screw would move rotated in the sense of description of C .

Note: If $\vec{F}$ is a conservative vector field, then $\nabla \times \vec{F} = \vec{0}$ and hence by Stoke's thorem,

$$\int_C \vec{F} \cdot d\vec{r} = 0.$$

Gauss's Divergence Theorem

If $\vec{F}$ is a continuously differentiable vector point function in a region V enclosed by a closed surface S , then

$$\int_S \vec{F} \cdot \hat{n} dS = \int_V \nabla \cdot \vec{F} dV,$$

where $\hat{n}$ is the outward drawn unit normal vector to the surface S .

Applications of Green's Theorem in a Plane

Example 1. A vector field $\vec{F}$ is given by $\vec{F} = (\sin y)\hat{i} + x(1 + \cos y)\hat{j}$. Evaluate the line integral $\int_C \vec{F} \cdot d\vec{r}$, where C is the circular path given by $x^2 + y^2 = a^2$.

Solution. Given vector field $\vec{F} = (\sin y)\hat{i} + x(1 + \cos y)\hat{j}$.

$\therefore$ Required line integral is

$$\begin{aligned} \int_C \vec{F} \cdot d\vec{r} &= \int_C \{(\sin y)\hat{i} + x(1 + \cos y)\hat{j}\} \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k}) \\ &= \int_C [(\sin y)dx + x(1 + \cos y)dy] \\ &= \int_C (Mdx + Ndy), \end{aligned} \quad \dots\dots\dots (1)$$

where $M = \sin y$ and $N = x(1 + \cos y)$. \dots\dots\dots (2)

Using Green's theorem, we have

$$\begin{aligned}\int_C (Mdx + Ndy) &= \iint_S \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dxdy \\ &= \iint_S [(1 + \cos y) - \cos y] dxdy, \text{ by (2)} \\ &= \iint_S dxdy = \text{area of the circle of radius } a \\ &= \pi a^2.\end{aligned}$$

Putting this value in (1), we get

$$\int_C \vec{F} \cdot d\vec{r} = \pi a^2. \quad \underline{\text{Ans.}}$$

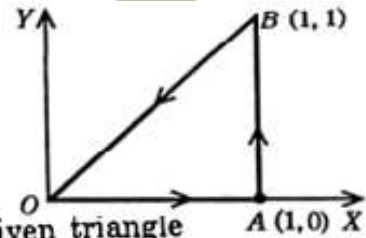
Example 2. Using Green's theorem, evaluate

$$\oint_C (x^2 y dx + x^2 dy),$$

where C is the boundary (described counter-clockwise) of the triangle with vertices (0, 0), (1, 0) and (1, 1).

Solution. By Green's theorem, we have

$$\begin{aligned}\int_C (M dx + N dy) &= \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy. \\ \therefore \int_C (x^2 y dx + x^2 dy) &= \iint_R (2x - x^2) dx dy, \\ &\text{where } R \text{ is the region of the given triangle}\end{aligned}$$



$$\begin{aligned}&= \int_0^1 (2x - x^2) \int_0^x dy dx = \int_0^1 (2x - x^2) [y]_0^x dx = \int_0^1 (2x - x^2) (x) dx \\ &= \int_0^1 (2x^2 - x^3) dx = \left[\frac{2x^3}{3} - \frac{x^4}{4} \right]_0^1 = \left(\frac{2}{3} - \frac{1}{4} \right) = \frac{5}{12}\end{aligned}$$

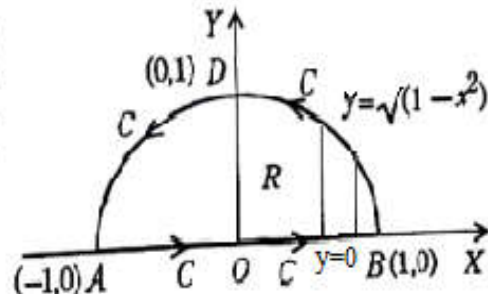
Example 3. Apply Green's theorem in the plane to evaluate the integral

$$\int_C [(2x^2 - y^2)dx + (x^2 + y^2)dy],$$

where C is the boundary of the surface enclosed by the x-axis and the semi-circle $y = \sqrt{1 - x^2}$.

Solution. Here C is the closed curve traversed in the positive direction by the straight line AOB and the semi-circle BDA. Also R is the region bounded by this curve C.

$$\text{We have } \int_C [(2x^2 - y^2) dx + (x^2 + y^2) dy]$$

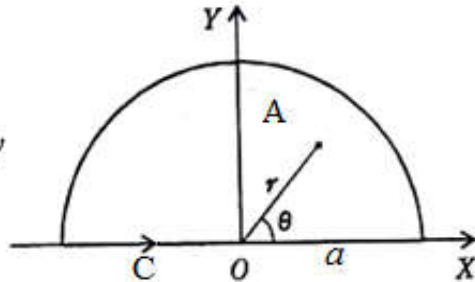


$$\begin{aligned}
&= \int_C M dx + N dy, \text{ where } M = 2x^2 - y^2, N = x^2 + y^2 \\
&= \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy, \text{ by Green's theorem} \\
&= \iint_R \left[\frac{\partial}{\partial x} (x^2 + y^2) - \frac{\partial}{\partial y} (2x^2 - y^2) \right] dx dy = \iint_R (2x + 2y) dx dy \\
&= \int_{x=-1}^1 \int_{y=0}^{\sqrt{1-x^2}} 2(x+y) dx dy, \text{ since for the region } R, y \text{ varies from} \\
&\quad 0 \text{ to } \sqrt{1-x^2} \text{ and } x \text{ varies from } -1 \text{ to } 1 \\
&= 2 \int_{x=-1}^1 \left[xy + \frac{y^2}{2} \right]_{y=0}^{\sqrt{1-x^2}} dx = 2 \int_{-1}^1 \left[x \sqrt{1-x^2} + \frac{1}{2} (1-x^2) \right] dx \\
&= 2 \int_0^1 (1-x^2) dx \quad \left[\because \int_{-1}^1 x \sqrt{1-x^2} dx = 0 \text{ and } \int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx \text{ if } f(-x) = f(x) \right] \\
&= 2 \left[x - \frac{x^3}{3} \right]_0^1 = 2 \left(1 - \frac{1}{3} \right) = \frac{4}{3}.
\end{aligned}$$

Example 4. Apply Green's theorem to evaluate $\oint_C [(2x^2 - y^2)dx + (x^2 + y^2)dy]$, where C is the boundary of the area enclosed by the x -axis and the upper half of the circle $x^2 + y^2 = a^2$.

Solution. By Green's theorem, we have

$$\begin{aligned}
&\int_C [(2x^2 - y^2) dx + (x^2 + y^2) dy] \\
&= \iint_A \left[\frac{\partial}{\partial x} (x^2 + y^2) - \frac{\partial}{\partial y} (2x^2 - y^2) \right] dx dy \\
&= 2 \iint_A (x + y) dx dy, \text{ where } A \text{ is the} \\
&\quad \text{region shown in the figure} \\
&= 2 \int_0^a \int_0^\pi r (\cos \theta + \sin \theta) \cdot r d\theta dr \quad [\text{Changing to polar coordinates.}] \\
&\quad \text{Here } r \text{ varies from } 0 \text{ to } a \text{ and } \theta \text{ varies from } 0 \text{ to } \pi.] \\
&= 2 \int_0^a r^2 dr \int_0^\pi (\cos \theta + \sin \theta) d\theta = 2 \cdot \frac{a^3}{3} \cdot (1 + 1) = \frac{4a^3}{3}.
\end{aligned}$$



Example 5. Verify Green's theorem in a plane for $\oint_C [(xy + y^2)dx + x^2 dy]$, where C is the closed curve of the region bounded by $y = x$ and $y = x^2$.

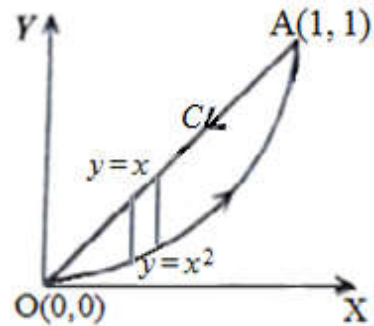
Solution. By Green's theorem in xy -plane, we have

$$\oint_C (Mdx + Ndy) = \iint_S \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dxdy, \quad \dots\dots\dots (1)$$

where C is the closed curve of the region S bounded by $y = x$ and $y = x^2$.

Let $M = xy + y^2$ and $N = x^2$.

The curves $y = x$ and $y = x^2$ intersect at $(0, 0)$ and $(1, 1)$. The positive direction in traversing C is as shown in the figure.



$$\text{R.H.S. of (1)} = \iint_S \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dxdy$$

$$\begin{aligned} &= \iint_S \left[\frac{\partial}{\partial x} (x^2) - \frac{\partial}{\partial y} (xy + y^2) \right] dxdy = \iint_S (2x - x - 2y) dxdy = \iint_S (x - 2y) dxdy \\ &= \int_{x=0}^1 \int_{y=x^2}^x (x - 2y) dy dx = \int_{x=0}^1 [xy - y^2]_{y=x^2}^x dx \\ &= \int_0^1 [x^2 - x^2 - x^3 + x^4] dx = \int_0^1 (x^4 - x^3) dx = \left[\frac{x^5}{5} - \frac{x^4}{4} \right]_0^1 = \frac{1}{5} - \frac{1}{4} = -\frac{1}{20}. \end{aligned}$$

Now let us evaluate the line integral along C. Along $y = x^2$, $dy = 2x dx$. Therefore along $y = x^2$, the line integral equals

$$\int_0^1 [(x)(x^2) + x^4] dx + x^2 (2x) dx = \int_0^1 (3x^3 + x^4) dx = \frac{19}{20}.$$

Along $y = x$, $dy = dx$. Therefore along $y = x$, the line integral equals

$$\int_1^0 [(x)(x) + x^2] dx + x^2 dx = \int_1^0 3x^2 dx = -1.$$

Therefore the required line integral $= \frac{19}{20} - 1 = -\frac{1}{20}$. Hence the theorem.

Example 6. Verify Green's theorem in a plane for

$$\oint_C [(3x^2 - 8y^2)dx + (4y - 6xy)dy],$$

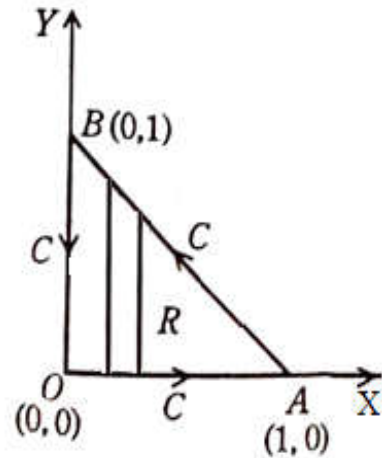
where C is the boundary of the region bounded by $x = 0$, $y = 0$ and $x + y = 1$.

Solution. By Green's theorem in xy -plane, we have

$$\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dxdy = \oint_C (Mdx + Ndy).$$

Here $M = 3x^2 - 8y^2$, $N = 4y - 6xy$.

The closed curve C consists of the straight line OA , the straight line AB and the straight line BO . The positive direction in traversing C is as shown in the figure and R is the region bounded by C .



$$\begin{aligned}
 \text{We have } \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy &= \iint_R \left[\frac{\partial}{\partial x} (4y - 6xy) - \frac{\partial}{\partial y} (3x^2 - 8y^2) \right] dx dy \\
 &= \iint_R [-6y + 16y] dx dy = 10 \iint_R y dx dy \\
 &= 10 \int_{x=0}^1 \int_{y=0}^{1-x} y dx dy, \because \text{for the region R, } x \text{ varies from 0 to 1} \\
 &\quad \text{and } y \text{ varies from 0 to } 1-x \\
 &= 10 \int_0^1 \left[\frac{y^2}{2} \right]_{y=0}^{1-x} dx, \text{ integrating with respect to } y \text{ regarding } x \text{ as const.} \\
 &= 5 \int_0^1 (1-x)^2 dx = 5 \left[\frac{(1-x)^3}{3(-1)} \right]_0^1 = 5 \left[0 - \frac{1}{(-3)} \right] = \frac{5}{3}. \quad \dots\dots\dots (1)
 \end{aligned}$$

Now let us evaluate the line integral along the curve C .

Along the straight line OA , we have $y = 0$, $dy = 0$ and x varies from 0 to 1.

$$\therefore \text{ line integral along } OA = \int_0^1 3x^2 dx = \left[x^3 \right]_0^1 = 1.$$

Along the straight line AB , we have $x = 1 - y$, $dx = -dy$ and y varies from 0 to 1.

$\therefore$ line integral along AB

$$\begin{aligned}
 &= \int_0^1 \{ [3(1-y)^2 - 8y^2] (-dy) + [4y - 6y(1-y)] dy \} \\
 &= \int_0^1 [-3(1-2y+y^2) + 8y^2 + 4y - 6y + 6y^2] dy \\
 &= \int_0^1 (11y^2 + 4y - 3) dy = \left[\frac{11}{3} y^3 + 2y^2 - 3y \right]_0^1 = \frac{11}{3} + 2 - 3 = \frac{8}{3}.
 \end{aligned}$$

Along the straight line BO , we have $x = 0$, $dx = 0$ and y varies from 1 to 0.

$$\therefore \text{line integral along } BO = \int_1^0 4y \, dy = 2 \left[y^2 \right]_1^0 = -2.$$

$$\therefore \text{total line integral along the closed curve } C = 1 + \frac{8}{3} - 2 = \frac{5}{3} \quad \dots\dots\dots (2)$$

Hence from (1) and (2), Green's theorem is verified.

Example 7. Apply Green's theorem in the plane to evaluate $\int_C \{(y - \sin x)dx + \cos x \, dy\}$, where C is the triangle enclosed by the lines $y = 0$, $x = \pi/2$, $y = 2x$.

Solution. Here C is the closed curve traversed in positive direction by ΔOAB and R is the region bounded by this curve C .

$$\text{We have } \int_C \{(y - \sin x) \, dx + \cos x \, dy\}$$

$$= \int_C (M \, dx + N \, dy),$$

$$\text{where } M = y - \sin x, \, N = \cos x$$

$$= \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx \, dy, \text{ by Green's theorem}$$

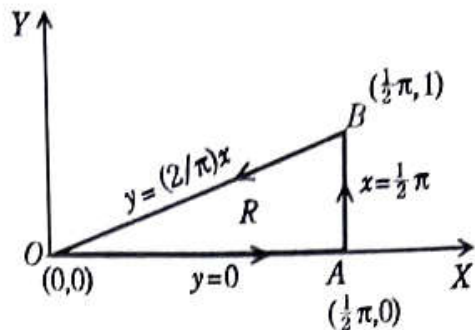
$$= \iint_R \left[\frac{\partial}{\partial x} (\cos x) - \frac{\partial}{\partial y} (y - \sin x) \right] dx \, dy = \int_{x=0}^{\pi/2} \int_{y=0}^{(2/\pi)x} (-\sin x - 1) \, dx \, dy$$

$\because$ for the region R , y varies from 0 to $(2/\pi)x$ and x varies from 0 to $\pi/2$

$$= \int_{x=0}^{\pi/2} \left[-y \sin x - y \right]_{y=0}^{(2/\pi)x} dx = \int_0^{\pi/2} \left[-\frac{2}{\pi} x \sin x - \frac{2}{\pi} x \right] dx$$

$$= -\frac{2}{\pi} \int_0^{\pi/2} (x + x \sin x) \, dx = -\frac{2}{\pi} \left[\frac{x^2}{2} \right]_0^{\pi/2} - \frac{2}{\pi} \left[x (-\cos x) \right]_0^{\pi/2} - \frac{2}{\pi} \int_0^{\pi/2} \cos x \, dx$$

$$= -\frac{2}{\pi} \cdot \frac{\pi^2}{8} - \frac{2}{\pi} \cdot 1 = -\frac{\pi}{4} - \frac{2}{\pi}.$$



Example 8. Use Green's theorem to evaluate $\int_C [(x^2 + xy)dx + (x^2 + y^2)dy]$,

where C is the square formed by the lines $y = \pm 1$, $x = \pm 1$.

Solution. By Green's theorem, we have $\int_C (Mdx + Ndy) = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx \, dy$.

$$\begin{aligned}\therefore \int_C (x^2 + xy) dx + (x^2 + y^2) dy &= \int_{-1}^1 \int_{-1}^1 \left[\frac{\partial}{\partial x} (x^2 + y^2) - \frac{\partial}{\partial y} (x^2 + xy) \right] dx dy \\ &= \int_{-1}^1 \int_{-1}^1 (2x - x) dx dy = \int_{-1}^1 \int_{-1}^1 x dx dy = \int_{-1}^1 x dx \int_{-1}^1 dy = \left[\frac{x^2}{2} \right]_{-1}^1 \times [y]_{-1}^1 = 0.\end{aligned}$$

Example 9. Show that the area bounded by a simple closed curve C is given by

$$\frac{1}{2} \oint_C (x dy - y dx).$$

Hence find the area of the ellipse $x = a \cos \theta$, $y = b \sin \theta$.

Solution. By Green's theorem in xy -plane, if R is a plane region bounded by a simple closed curve C, then

$$\oint_C (M dx + N dy) = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy. \quad \dots\dots\dots (1)$$

Putting $M = -y$ and $N = x$ in (1), we get

$$\oint_C (M dx + N dy) = \iint_R \left(\frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} \right) dx dy = 2 \iint_R dx dy = 2A,$$

where A is the area bounded by C.

$$\text{Hence } A = \frac{1}{2} \oint_C (x dy - y dx). \quad \underline{\text{Proved.}}$$

Now the area of the ellipse $x = a \cos \theta$, $y = b \sin \theta$ is

$$\begin{aligned}\text{Hence } A &= \frac{1}{2} \oint_C (x dy - y dx) = \frac{1}{2} \oint_{\theta=0}^{2\pi} (a \cos \theta db \sin \theta - b \sin \theta da \cos \theta) \\ &= \frac{1}{2} ab \oint_{\theta=0}^{2\pi} (\cos^2 \theta + \sin^2 \theta) d\theta = \frac{1}{2} ab \oint_{\theta=0}^{2\pi} d\theta = \frac{1}{2} ab 2\pi \\ &= \pi ab. \quad \underline{\text{Ans.}}\end{aligned}$$

Example 10. Use Green's theorem to evaluate the integral

$$\oint_C (-y^3 dx + x^3 dy), \text{ where C is the circle } x^2 + y^2 = 1.$$

Solution. Try yourself.

Example 11. Verify Green's theorem for $\oint_C [(3x^2 - 8y^2)dx + (4y - 6xy)dy]$,

where C is the region bounded by the parabolas $y = \sqrt{x}$ and $y = x^2$.

Solution. Try yourself.

LECTURE NO.:- 39

Applications of Gauss's Divergence Theorem

Ex. 1. For any closed surface S , prove that $\iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} dS = 0$.

Solution. By divergence theorem, we have

$$\begin{aligned} \iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} dS &= \iiint_V (\text{div curl } \mathbf{F}) dV, \text{ where } V \text{ is the volume enclosed by } S \\ &= 0, \text{ since } \text{div curl } \mathbf{F} = 0. \end{aligned}$$

Ex. 2. If $\mathbf{F} = ax\mathbf{i} + by\mathbf{j} + cz\mathbf{k}$, a, b, c are constants show that

$$\iint_S \mathbf{F} \cdot \mathbf{n} dS = \frac{4}{3}\pi(a+b+c), \text{ where } S \text{ is the surface of a unit sphere.}$$

Solution. By the divergence theorem, we have

$$\begin{aligned} \iint_S \mathbf{F} \cdot \mathbf{n} dS &= \iiint_V (\nabla \cdot \mathbf{F}) dV, \text{ where } V \text{ is the volume enclosed by } S \\ &= \iiint_V [\nabla \cdot (ax\mathbf{i} + by\mathbf{j} + cz\mathbf{k})] dV = \iiint_V \left[\frac{\partial}{\partial x}(ax) + \frac{\partial}{\partial y}(by) + \frac{\partial}{\partial z}(cz) \right] dV \\ &= \iiint_V (a+b+c) dV = (a+b+c)V = (a+b+c)\frac{4}{3}\pi, \end{aligned}$$

since the volume V enclosed by a sphere of unit radius is equal to $\frac{4}{3}\pi(1)^3$ i.e., $\frac{4}{3}\pi$.

Ex. 3. Prove that $\iiint_V \frac{dV}{r^2} = \iint_S \frac{\mathbf{r} \cdot \mathbf{n}}{r^2} dS$.

Solution. We have

$$\begin{aligned} \iint_S \frac{\mathbf{r} \cdot \mathbf{n}}{r^2} dS &= \iint_S \left(\frac{\mathbf{r}}{r^2} \right) \cdot \mathbf{n} dS = \iiint_V \nabla \cdot \left(\frac{\mathbf{r}}{r^2} \right) dV, \text{ by divergence theorem.} \\ \text{Now } \nabla \cdot \left(\frac{\mathbf{r}}{r^2} \right) &= \frac{1}{r^2} (\nabla \cdot \mathbf{r}) + \mathbf{r} \cdot \nabla \left(\frac{1}{r^2} \right) = \frac{3}{r^2} + \mathbf{r} \cdot \left(-\frac{2}{r^3} \nabla r \right) = \frac{3}{r^2} - \frac{2}{r^3} \left(\mathbf{r} \cdot \frac{\mathbf{r}}{r} \right) \\ &= \frac{3}{r^2} - \frac{2}{r^4} r^2 = \frac{1}{r^2}. \end{aligned}$$

Hence $\iiint_V \frac{dV}{r^2} = \iint_S \frac{\mathbf{r} \cdot \mathbf{n}}{r^2} dS$. Proved.

Ex. 4. If $\mathbf{F} = \nabla\phi$, $\nabla^2\phi = -4\pi\rho$, show that $\iint_S \mathbf{F} \cdot \mathbf{n} dS = -4\pi \iiint_V \rho dV$.

Solution. By divergence theorem, we have $\iint_S \mathbf{F} \cdot \mathbf{n} dS = \iiint_V (\nabla \cdot \mathbf{F}) dV$.

Now $\nabla \cdot \mathbf{F} = \nabla \cdot (\nabla\phi) = \nabla^2\phi = -4\pi\rho$.

$$\therefore \iint_S \mathbf{F} \cdot \mathbf{n} dS = \iiint_V (-4\pi\rho) dV = -4\pi \iiint_V \rho dV.$$

Ex. 5. If ϕ is harmonic in V , then $\iint_S \phi \frac{\partial \phi}{\partial n} dS = \iiint_V |\nabla \phi|^2 dV$.

Solution. We have $\iint_S \phi \frac{\partial \phi}{\partial n} dS = \iint_S \left(\phi \frac{\partial \phi}{\partial n} \mathbf{n} \right) \cdot \mathbf{n} dS$
 $= \iint_S (\phi \nabla \phi) \cdot \mathbf{n} dS = \iiint_V \nabla \cdot (\phi \nabla \phi) dV$, by divergence theorem
 $= \iiint_V [(\nabla \phi \cdot \nabla \phi) + \phi (\nabla \cdot \nabla \phi)] dV = \iiint_V [(\nabla \phi)^2 + \phi \nabla^2 \phi] dV$
 $= \iiint_V |\nabla \phi|^2 dV$, since $\nabla^2 \phi = 0$ and $(\nabla \phi)^2 = |\nabla \phi|^2$.

Ex. 6. Use the divergence theorem to show that $\iint_S \nabla (x^2 + y^2 + z^2) \cdot dS = 6V$

where S is any closed surface enclosing volume V .

Solution. We have $\nabla (x^2 + y^2 + z^2) = \left(\mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z} \right) (x^2 + y^2 + z^2)$
 $= 2x \mathbf{i} + 2y \mathbf{j} + 2z \mathbf{k} = 2(x\mathbf{i} + y\mathbf{j} + z\mathbf{k})$.

$$\therefore \iint_S \nabla (x^2 + y^2 + z^2) \cdot dS = \iint_S \nabla (x^2 + y^2 + z^2) \cdot \mathbf{n} dS,$$

where $\mathbf{n}$ is a outward drawn unit normal vector to S

$$= \iint_S 2(x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) \cdot \mathbf{n} dS = 2 \iint_S \text{div}(x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) dV,$$

[By Divergence theorem]

where V is the volume enclosed by S

$$= 2 \iiint_V \left(\mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z} \right) \cdot (x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) dV$$

$$= 2 \iiint_V \left(\frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z} \right) dV = 2 \iiint_V 3 dV = 6 \iiint_V dV = 6V.$$

Ex. 7. Using the divergence theorem, show that the volume V of a region T bounded by a surface S is $= \frac{1}{3} \iint_S (x dy dz + y dz dx + z dx dy)$.

Solution. By divergence theorem, we have

$$\iint_S x dy dz = \iiint_V \left(\frac{\partial}{\partial x} (x) \right) dV = \iiint_V dV = V,$$

$$\iint_S y dz dx = \iiint_V \left(\frac{\partial}{\partial y} (y) \right) dV = \iiint_V dV = V,$$

$$\iint_S z dx dy = \iiint_V \left(\frac{\partial}{\partial z} (z) \right) dV = \iiint_V dV = V.$$

Adding these results, we get

$$3V = \iint_S (x dy dz + y dz dx + z dx dy) \quad \text{or} \quad V = \frac{1}{3} \iint_S (x dy dz + y dz dx + z dx dy).$$

Ex. 8. Use the divergence theorem to evaluate $\iint_S \vec{A} \cdot d\vec{S}$,
where $\vec{A} = x^3 \mathbf{i} + y^3 \mathbf{j} + z^3 \mathbf{k}$ and S is the surface of the sphere $x^2 + y^2 + z^2 = a^2$.

Solution. If V is the volume of the sphere $x^2 + y^2 + z^2 = a^2$,
then by divergence theorem, we have

$$\begin{aligned} \iint_S \vec{A} \cdot d\vec{S} &= \iiint_V \operatorname{div} \vec{A} dV \\ &= \iiint_V \left(\mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z} \right) \cdot (x^3 \mathbf{i} + y^3 \mathbf{j} + z^3 \mathbf{k}) dV \\ &= \iiint_V (3x^2 + 3y^2 + 3z^2) dV = 3 \iiint_V (x^2 + y^2 + z^2) dV \\ &= 3 \times 8 \int_0^{\pi/2} \int_0^{\pi/2} \int_0^a r^2 (r^2 \sin \theta dr d\theta d\phi), \\ &\quad \text{putting } x = r \sin \theta \cos \phi, y = r \sin \theta \sin \phi, z = r \cos \theta \\ &\quad \text{and } dV = (dr)(r d\theta)(r \sin \theta d\phi) = r^2 \sin \theta dr d\theta d\phi \\ &= 3 \times 8 \int_0^{\pi/2} d\phi \int_0^{\pi/2} \sin \theta d\theta \int_0^a r^4 dr \\ &= 24 \left[\phi \right]_0^{\pi/2} \times \left[-\cos \theta \right]_0^{\pi/2} \times \left[\frac{r^5}{5} \right]_0^a = 24 \left(\frac{\pi}{2} \right) (-0 + 1) \left(\frac{a^5}{5} \right) = \frac{12\pi a^5}{5}. \end{aligned}$$

Ex. 9. Verify divergence theorem for $\mathbf{F} = (x^2 - yz) \mathbf{i} + (y^2 - zx) \mathbf{j} + (z^2 - xy) \mathbf{k}$ taken over the rectangular parallelepiped $0 \leq x \leq a, 0 \leq y \leq b, 0 \leq z \leq c$.

Solution. We have $\operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F}$

$$= \frac{\partial}{\partial x} (x^2 - yz) + \frac{\partial}{\partial y} (y^2 - zx) + \frac{\partial}{\partial z} (z^2 - xy) = 2x + 2y + 2z.$$

$$\therefore \text{volume integral} = \iiint_V \nabla \cdot \mathbf{F} dV = \iiint_V 2(x + y + z) dV$$

$$= 2 \int_{z=0}^c \int_{y=0}^b \int_{x=0}^a (x + y + z) dx dy dz = 2 \int_{z=0}^c \int_{y=0}^b \left[\frac{x^2}{2} + yx + zx \right]_{x=0}^a dy dz$$

$$= 2 \int_{z=0}^c \int_{y=0}^b \left[\frac{a^2}{2} + ay + az \right] dy dz = 2 \int_{z=0}^c \left[\frac{a^2}{2} y + a \frac{y^2}{2} + azy \right]_{y=0}^b dz$$

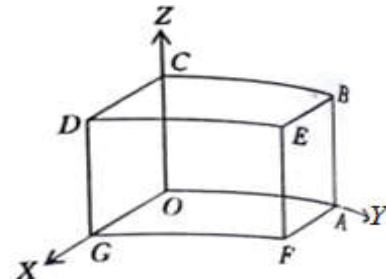
$$= 2 \int_{z=0}^c \left[\frac{a^2 b}{2} + \frac{ab^2}{2} + abz \right] dz = 2 \left[\frac{a^2 b}{2} z + \frac{ab^2}{2} z + ab \frac{z^2}{2} \right]_0^c$$

$$= [a^2 bc + ab^2 c + abc^2] = abc(a + b + c).$$

Surface Integral. We shall now calculate

$$\iint_S \mathbf{F} \cdot \mathbf{n} dS$$

over the six faces of the rectangular parallelepiped.
Over the face $DEFG$, $\mathbf{n} = \mathbf{i}$, $x = a$.



Therefore, $\iint_{DEFG} \mathbf{F} \cdot \mathbf{n} dS$

$$= \int_{z=0}^c \int_{y=0}^b [(a^2 - yz)\mathbf{i} + (y^2 - za)\mathbf{j} + (z^2 - ay)\mathbf{k}] \cdot \mathbf{i} dy dz$$

$$= \int_{z=0}^c \int_{y=0}^b (a^2 - yz) dy dz = \int_{z=0}^c \left[a^2 y - z \frac{y^2}{2} \right]_{y=0}^b dz$$

$$= \int_{z=0}^c \left[a^2 b - \frac{zb^2}{2} \right] dz = \left[a^2 bz - \frac{z^2}{4} b^2 \right]_0^c = a^2 bc - \frac{c^2 b^2}{4}.$$

Over the face $ABCO$, $\mathbf{n} = -\mathbf{i}$, $x = 0$. Therefore

$$\iint_{ABCO} \mathbf{F} \cdot \mathbf{n} dS = \iint (0 - yz)\mathbf{i} + \dots + \dots \cdot (-\mathbf{i}) dy dz$$

$$= \int_{z=0}^c \int_{y=0}^b yz dy dz = \int_{z=0}^c \left[\frac{y^2}{2} z \right]_{y=0}^b dz = \int_{z=0}^c \frac{b^2}{2} z dz = \frac{b^2}{2} \frac{c^2}{2}.$$

Over the face $ABEF$, $\mathbf{n} = \mathbf{j}$, $y = b$. Therefore

$$\iint_{ABEF} \mathbf{F} \cdot \mathbf{n} dS = \int_{z=0}^c \int_{x=0}^a [(x^2 - bz)\mathbf{i} + (b^2 - zx)\mathbf{j} + (z^2 - bx)\mathbf{k}] \cdot \mathbf{j} dx dz$$

$$= \int_{z=0}^c \int_{x=0}^a (b^2 - zx) dx dz = b^2 ca - \frac{a^2 c^2}{4}.$$

Over the face $OGDC$, $\mathbf{n} = -\mathbf{j}$, $y = 0$. Therefore

$$\iint_{OGDC} \mathbf{F} \cdot \mathbf{n} dS = \int_{z=0}^c \int_{x=0}^a zx dx dz = \frac{c^2 a^2}{4}.$$

Over the face $BCDE$, $\mathbf{n} = \mathbf{k}$, $z = c$.

Therefore $\iint_{BCDE} \mathbf{F} \cdot \mathbf{n} dS = \int_{y=0}^b \int_{x=0}^a (c^2 - xy) dx dy = c^2 ab - \frac{a^2 b^3}{4}.$

Over the face $AFGO$, $\mathbf{n} = -\mathbf{k}$, $z = 0$.

Therefore $\iint_{AFGO} \mathbf{F} \cdot \mathbf{n} dS = \int_{y=0}^b \int_{x=0}^a xy dx dy = \frac{a^2 b^2}{4}.$

Adding the six surface integrals, we get

$$\iint_S \mathbf{F} \cdot \mathbf{n} dS = \left(a^2 bc - \frac{c^2 b^2}{4} + \frac{b^2 c^2}{4} \right) + \left(b^2 ca - \frac{a^2 c^2}{4} + \frac{a^2 c^2}{4} \right) + \left(c^2 ab - \frac{a^2 b^3}{4} + \frac{a^2 b^3}{4} \right)$$

$$= abc(a + b + c).$$

Hence the theorem is verified.

Ex. 10. Find the value of $\iint_S (\mathbf{F} \times \nabla \phi) \cdot \mathbf{n} dS$.

where $\mathbf{F} = x^2 \mathbf{i} + y^2 \mathbf{j} + z^2 \mathbf{k}$, $\phi = xy + yz + zx$, S is $x = \pm 1, y = \pm 1, z = \pm 1$.

Solution. By Gauss divergence theorem,

$$\iint_S (\mathbf{F} \times \nabla \phi) \cdot \mathbf{n} dS = \iiint_V \nabla \cdot (\mathbf{F} \times \nabla \phi) dV,$$

where V is the volume enclosed by the surface S .

Now from a vector identity, we know that $\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B})$.

$$\therefore \nabla \cdot (\mathbf{F} \times \nabla \phi) = \nabla \phi \cdot (\nabla \times \mathbf{F}) - \mathbf{F} \cdot (\nabla \times (\nabla \phi))$$

$$= \nabla \phi \cdot (\nabla \times \mathbf{F}).$$

$$[\because \nabla \times (\nabla \phi) = \text{curl grad } \phi = \mathbf{0}]$$

$$\text{Now } \nabla \times \mathbf{F} = \nabla \times (x^2 \mathbf{i} + y^2 \mathbf{j} + z^2 \mathbf{k}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 & y^2 & z^2 \end{vmatrix}$$

$$= \left[\frac{\partial}{\partial y} (z^2) - \frac{\partial}{\partial z} (y^2) \right] \mathbf{i} + \left[\frac{\partial}{\partial z} (x^2) - \frac{\partial}{\partial x} (z^2) \right] \mathbf{j} + \left[\frac{\partial}{\partial x} (y^2) - \frac{\partial}{\partial y} (x^2) \right] \mathbf{k}$$

$$= 0\mathbf{i} + 0\mathbf{j} + 0\mathbf{k} = \mathbf{0}.$$

$$\therefore \nabla \cdot (\mathbf{F} \times \nabla \phi) = (\nabla \phi) \cdot \mathbf{0} = 0.$$

$$\text{Hence the given integral} = \iiint_V 0 dV = 0.$$

Ex. 11. Verify divergence theorem for $\mathbf{F} = (2x - z) \mathbf{i} + x^2 y \mathbf{j} - xz^2 \mathbf{k}$ taken over the region bounded by $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$.

Solution. By divergence theorem, we have $\iint_S \mathbf{F} \cdot \mathbf{n} dS = \iiint_V (\nabla \cdot \mathbf{F}) dV$,

where S is the surface of the cube bounded by the planes $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$ and V is the volume enclosed by the surface S .

We have $\nabla \cdot \mathbf{F} = \nabla \cdot [(2x - z) \mathbf{i} + x^2 y \mathbf{j} - xz^2 \mathbf{k}]$

$$= \frac{\partial}{\partial x} (2x - z) + \frac{\partial}{\partial y} (x^2 y) + \frac{\partial}{\partial z} (-xz^2) = 2 + x^2 - 2xz.$$

$$\therefore \iiint_V (\nabla \cdot \mathbf{F}) dV = \iiint_V (2 + x^2 - 2xz) dV = \int_{z=0}^1 \int_{y=0}^1 \int_{x=0}^1 (2 + x^2 - 2xz) dx dy dz$$

$$= \int_{z=0}^1 \int_{y=0}^1 \left[2x + \frac{x^3}{3} - x^2 z \right]_{x=0}^1 dy dz = \int_{z=0}^1 \int_{y=0}^1 \left(2 + \frac{1}{3} - z \right) dy dz$$

$$= \int_{z=0}^1 \int_{y=0}^1 \left(\frac{7}{3} - z \right) dy dz = \int_{z=0}^1 \left[\frac{7}{3} y - zy \right]_{y=0}^1 dz = \int_0^1 \left(\frac{7}{3} - z \right) dz$$

$$= \left[\frac{7}{3} z - \frac{1}{2} z^2 \right]_0^1 = \frac{7}{3} - \frac{1}{2} = \frac{11}{6}. \quad \dots(1)$$

We shall now calculate $\iint_S \mathbf{F} \cdot \mathbf{n} dS$ over the six faces of the cube.

Draw figure as in solved example 9. Over the face $OABC$ which lies in the yz -plane, $\mathbf{n} = -\mathbf{i}$, $x = 0$. Therefore

$$\begin{aligned} \iint_{OABC} \mathbf{F} \cdot \mathbf{n} dS &= \int_{z=0}^1 \int_{y=0}^1 (-z\mathbf{i}) \cdot (-\mathbf{i}) dy dz \\ &= \int_{z=0}^1 \int_{y=0}^1 z dy dz = \int_{z=0}^1 z [y]_{y=0}^1 dz = \int_0^1 z dz = \left[\frac{z^2}{2} \right]_0^1 = \frac{1}{2}. \end{aligned}$$

Over the opposite face $DEFG$, $x = 1$, $\mathbf{n} = \mathbf{i}$.

$$\begin{aligned}\text{Therefore } \iint_{DEFG} \mathbf{F} \cdot \mathbf{n} dS &= \int_{z=0}^1 \int_{y=0}^1 [(2-z)\mathbf{i} + y\mathbf{j} - z^2\mathbf{k}] \cdot \mathbf{i} dy dz \\ &= \int_{z=0}^1 \int_{y=0}^1 (2-z) dy dz = \int_{z=0}^1 (2-z) [y]_{y=0}^1 dz = \int_0^1 (2-z) dz = \left[2z - \frac{z^2}{2} \right]_0^1 = 2 - \frac{1}{2} = \frac{3}{2}\end{aligned}$$

Over the face $OGDC$ which lies in the zx -plane, $y = 0$, $\mathbf{n} = -\mathbf{j}$.

$$\therefore \iint_{OGDC} \mathbf{F} \cdot \mathbf{n} dS = \int_{z=0}^1 \int_{x=0}^1 [(2x-z)\mathbf{i} - xz^2\mathbf{k}] \cdot (-\mathbf{j}) dz dx = \int_{z=0}^1 \int_{x=0}^1 0 dz dx = 0$$

Over the opposite face $ABEF$, $y = 1$, $\mathbf{n} = \mathbf{j}$.

$$\begin{aligned}\therefore \iint_{ABEF} \mathbf{F} \cdot \mathbf{n} dS &= \int_{z=0}^1 \int_{x=0}^1 [(2x-z)\mathbf{i} + x^2\mathbf{j} - xz^2\mathbf{k}] \cdot \mathbf{j} dz dx = \int_{z=0}^1 \int_{x=0}^1 x^2 dz dx \\ &= \int_{z=0}^1 \left[\frac{x^3}{3} \right]_{x=0}^1 dz = \int_0^1 \frac{1}{3} dz = \frac{1}{3} [z]_0^1 = \frac{1}{3}\end{aligned}$$

Over the face $OGFA$ which lies in the xy -plane, $z = 0$, $\mathbf{n} = -\mathbf{k}$.

$$\therefore \iint_{OGFA} \mathbf{F} \cdot \mathbf{n} dS = \int_{x=0}^1 \int_{y=0}^1 (2x\mathbf{i} + x^2y\mathbf{j}) \cdot (-\mathbf{k}) dx dy = \int_{x=0}^1 \int_{y=0}^1 0 dx dy = 0$$

Over the opposite face $BCDE$, $z = 1$, $\mathbf{n} = \mathbf{k}$.

$$\begin{aligned}\therefore \iint_{BCDE} \mathbf{F} \cdot \mathbf{n} dS &= \int_{x=0}^1 \int_{y=0}^1 [(2x-1)\mathbf{i} + x^2y\mathbf{j} - x\mathbf{k}] \cdot \mathbf{k} dx dy \\ &= \int_{x=0}^1 \int_{y=0}^1 -x dx dy = \int_{x=0}^1 -x [y]_{y=0}^1 dx = \int_0^1 -x dx = -\left[\frac{x^2}{2} \right]_0^1 = -\frac{1}{2}\end{aligned}$$

Adding the six surface integrals, we get

$$\iint_S \mathbf{F} \cdot \mathbf{n} dS = \frac{1}{2} + \frac{3}{2} + 0 + \frac{1}{3} + 0 - \frac{1}{2} = \frac{11}{6} \quad \dots(2)$$

From (1) and (2), we see that $\iiint_V \text{div } \mathbf{F} dV = \iint_S \mathbf{F} \cdot \mathbf{n} dS$.

This verifies Gauss divergence theorem.

Ex. 12. Verify divergence theorem for $\mathbf{F} = 4xz\mathbf{i} - y^2\mathbf{j} + yz\mathbf{k}$ taken over the cube bounded by $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$.

Solution. Proceed as in above example 11. Here we shall have

$$\begin{aligned}\iiint_V (\nabla \cdot \mathbf{F}) dV &= \iiint_V \left[\frac{\partial}{\partial x} (4xz) + \frac{\partial}{\partial y} (-y^2) + \frac{\partial}{\partial z} (yz) \right] dV \\ &= \int_{z=0}^1 \int_{y=0}^1 \int_{x=0}^1 (4z - 2y + y) dx dy dz = \int_{z=0}^1 \int_{y=0}^1 (4z - y) dx dy dz = \frac{3}{2}\end{aligned}$$

The six surface integrals will come out to be 2, 0, -1, 0, $\frac{1}{2}$ and 0. Their sum = $\frac{3}{2}$.

Hence the theorem is verified.

Ex. 13. If S is the surface of the sphere $x^2 + y^2 + z^2 = 1$, evaluate

$$\int_S (x\hat{i} + 2y\hat{j} + 3z\hat{k}) \cdot \overrightarrow{dS}$$

Solution. Let $\vec{F} = x\hat{i} + 2y\hat{j} + 3z\hat{k}$ and $\hat{n}$ is the outward drawn unit normal vector to the surface S. Then the required integral is

$$\int_S (x\hat{i} + 2y\hat{j} + 3z\hat{k}) \cdot \vec{dS} = \int_S \vec{F} \cdot \hat{n} dS. \quad \dots\dots\dots (1)$$

Let V be the volume enclosed by the surface S. Then by Gauss's divergence theorem, we have

$$\int_S \vec{F} \cdot \hat{n} dS = \int_V \nabla \cdot \vec{F} dV \quad \dots\dots\dots (2)$$

$$\begin{aligned} \text{Now } \nabla \cdot \vec{F} &= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot (x\hat{i} + 2y\hat{j} + 3z\hat{k}) \\ &= \frac{\partial x}{\partial x} + \frac{\partial 2y}{\partial y} + \frac{\partial 3z}{\partial z} = 1 + 2 + 3 \\ &= 6. \end{aligned}$$

Putting this value in (2), we get

$$\begin{aligned} \int_S \vec{F} \cdot \hat{n} dS &= \int_V 6 dV \\ &= 6 \int_V dV = 6V = 6 \times \text{Volume enclosed by the surface S} \\ &= 6 \times \text{Volume of the sphere of radius 1} = 6 \times \frac{4}{3} \pi 1^3 \\ &= 8\pi. \end{aligned}$$

Putting this value in (1), we get

$$\int_S (x\hat{i} + 2y\hat{j} + 3z\hat{k}) \cdot \vec{dS} = 8\pi. \quad \underline{\text{Ans.}}$$

Ex. 14. Evaluate $\int_S \vec{F} \cdot \hat{n} dS$, where $\vec{F} = 2xy\hat{i} + yz^2\hat{j} + xz\hat{k}$ and S is the surface of the region bounded by $x = 0, y = 0, z = 0, y = 3$ and $x + 2z = 6$.

Solution. Try yourself.

Ex. 15. Find $\int_S \vec{F} \cdot \hat{n} dS$, where $\vec{F} = (2x + 3z)\hat{i} - (xz + y)\hat{j} + (y^2 + 2z)\hat{k}$ and S is the surface of the sphere having centre at $(3, -1, 2)$ and radius 3.

Solution. Try yourself.

Ex. 16. By transforming to a triple integral, evaluate

$$I = \iiint_S (x^3 dydz + x^2 y dzdx + x^2 z dxdy),$$

where S is the closed surface bounded by the planes $z = 0, z = b$ and the cylinder $x^2 + y^2 = a^2$.

Solution. Try yourself.

Ex. 17. Verify the divergence theorem for $\vec{F} = 4x\hat{i} - 2y^2\hat{j} + z^2\hat{k}$ taken over the region bounded by the surfaces $x^2 + y^2 = 4, z = 0, z = 3$.

Solution. Try yourself.

LECTURE NO.:- 40

Applications of Stoke's Theorem

Ex. 1. By Stoke's theorem, prove that $\text{curl grad } \phi = 0$.

Solution. Let S be any surface enclosed by a simple closed curve C . Then by Stoke's theorem, we have
$$\iint_S (\text{curl grad } \phi) \cdot \mathbf{n} \, dS = \oint_C \text{grad } \phi \cdot d\mathbf{r}. \quad \dots (1)$$

$$\text{Now grad } \phi \cdot d\mathbf{r} = \left(\frac{\partial \phi}{\partial x} \mathbf{i} + \frac{\partial \phi}{\partial y} \mathbf{j} + \frac{\partial \phi}{\partial z} \mathbf{k} \right) \cdot (dx \mathbf{i} + dy \mathbf{j} + dz \mathbf{k}) = \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy + \frac{\partial \phi}{\partial z} dz = d\phi.$$

$$\therefore \oint_C \text{grad } \phi \cdot d\mathbf{r} = \oint_C d\phi = \left[\phi \right]_A^A = 0, \text{ where } A \text{ is any point on } C.$$

Hence by (1), we have

$$\iint_S (\text{curl grad } \phi) \cdot \mathbf{n} \, dS = 0.$$

Now this equation is true for all surface elements of S .

Therefore we have $\text{curl grad } \phi = 0$.

Ex. 2. Verify Stoke's theorem for $\mathbf{F} = y\mathbf{i} + z\mathbf{j} + x\mathbf{k}$, where S is the upper half surface of the sphere $x^2 + y^2 + z^2 = 1$ and C is its boundary.

Solution. The boundary C of the surface S is a circle in the xy -plane, whose centre is origin and radius is unity (1). So the equations of C are $x^2 + y^2 = 1, z = 0$ and its parametric equations may be taken as $x = \cos t, y = \sin t, z = 0, 0 \leq t < 2\pi$. Then we have

$$\begin{aligned} \oint_C \mathbf{F} \cdot d\mathbf{r} &= \oint_C (y\mathbf{i} + z\mathbf{j} + x\mathbf{k}) \cdot (dx \mathbf{i} + dy \mathbf{j} + dz \mathbf{k}) \\ &= \oint_C (y \, dx + z \, dy + x \, dz) = \oint_C y \, dx, \text{ since on } C, z = 0 \text{ and } dz = 0 \\ &= \int_0^{2\pi} \sin t \frac{dx}{dt} dt = \int_0^{2\pi} -\sin^2 t \, dt \\ &= -\frac{1}{2} \int_0^{2\pi} (1 - \cos 2t) \, dt = -\frac{1}{2} \left[t - \frac{\sin 2t}{2} \right]_0^{2\pi} = -\pi. \end{aligned} \quad \dots (1)$$

Now let us evaluate $\iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} \, dS$.

$$\text{We have curl } \mathbf{F} = \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y & z & x \end{vmatrix} = -\mathbf{i} - \mathbf{j} - \mathbf{k}.$$

Let S_1 be the plane region bounded by the circle C . If S' is the surface consisting of the surfaces S and S_1 , then S' is a closed surface. Let V be the volume bounded by S' .

If $\mathbf{n}$ denotes the outward drawn (drawn outside the region V) unit normal vector to S' , then on the plane surface S_1 , we have $\mathbf{n} = -\mathbf{k}$. Note that $\mathbf{k}$ is a unit vector normal to S_1 drawn into the region V .

By Gauss divergence theorem, we have

$$\iint_{S'} (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS = \iiint_V [\text{div} (\text{curl } \mathbf{F})] \, dV = 0. \quad [\because \text{div} (\text{curl } \mathbf{F}) = 0]$$

$$\begin{aligned}
 \therefore \quad & \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS + \iint_{S_1} (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS = 0 \quad \because S' \text{ consists of } S \text{ and } S_1 \\
 \text{or} \quad & \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS - \iint_{S_1} (\text{curl } \mathbf{F}) \cdot \mathbf{k} \, dS = 0 \quad \because \text{on } S_1, \mathbf{n} = -\mathbf{k} \\
 \text{or} \quad & \iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} \, dS = \iint_{S_1} \text{curl } \mathbf{F} \cdot \mathbf{k} \, dS \\
 & = \iint_{S_1} (-\mathbf{i} - \mathbf{j} - \mathbf{k}) \cdot \mathbf{k} \, dS = \iint_{S_1} (-1) \, dS = - \iint_{S_1} dS = -S_1.
 \end{aligned}$$

But S_1 = area of a circle of radius 1 = $\pi(1)^2 = \pi$.

$$\therefore \quad \iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} \, dS = -\pi. \quad \dots (2)$$

Hence from (1) and (2), the theorem is verified.

Ex. 3. Verify Stoke's theorem for $\mathbf{F} = (2x - y)\mathbf{i} - yz^2\mathbf{j} - y^2z\mathbf{k}$, where S is the upper half surface of the sphere $x^2 + y^2 + z^2 = 1$ and C is its boundary.

Solution. The boundary C of S is a circle in the xy -plane of radius unity and centre origin. Suppose $x = \cos t$, $y = \sin t$, $z = 0$, $0 \leq t < 2\pi$ are parametric equations of

$$\begin{aligned}
 C. \text{ Then } \oint_C \mathbf{F} \cdot d\mathbf{r} &= \oint_C [(2x - y)\mathbf{i} - yz^2\mathbf{j} - y^2z\mathbf{k}] \cdot (dx\mathbf{i} + dy\mathbf{j} + dz\mathbf{k}) \\
 &= \oint_C [(2x - y) dx - yz^2 dy - y^2z dz] \\
 &= \oint_C (2x - y) dx, \text{ since } z = 0 \text{ and } dz = 0 \\
 &= \int_0^{2\pi} (2 \cos t - \sin t) \frac{dx}{dt} dt = - \int_0^{2\pi} (2 \cos t - \sin t)(-\sin t) dt \\
 &= - \int_0^{2\pi} [\sin 2t - \frac{1}{2}(1 - \cos 2t)] dt = - \left[-\frac{\cos 2t}{2} - \frac{1}{2}t + \frac{1}{2} \frac{\sin 2t}{2} \right]_0^{2\pi} \\
 &= - \left[\left(-\frac{1}{2} + \frac{1}{2}\right) - \frac{1}{2}(\pi - 0) + \frac{1}{4}(0 - 0) \right] = \pi. \quad \dots (1)
 \end{aligned}$$

$$\text{And } (\nabla \times \mathbf{F}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x - y & -yz^2 & -y^2z \end{vmatrix} = (-2yz + 2yz)\mathbf{i} - (0 - 0)\mathbf{j} + (0 + 1)\mathbf{k}$$

Let S_1 be the plane region bounded by the circle C . If S' is the surface of the surfaces S and S_1 , then S' is a closed surface.

$\therefore$ by an application of Gauss divergence theorem, we have

$$\iint_{S'} \text{curl } \mathbf{F} \cdot \mathbf{n} \, dS = 0 \quad [\text{See Ex. 1 of previous lecture}]$$

$$\begin{aligned}
 \text{or } \iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} \, dS + \iint_{S_1} \text{curl } \mathbf{F} \cdot \mathbf{n} \, dS &= 0 & [\because S' \text{ consists of } S \text{ and } S_1] \\
 \text{or } \iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} \, dS - \iint_{S_1} \text{curl } \mathbf{F} \cdot \mathbf{k} \, dS &= 0 & [\because \text{ on } S_1, \mathbf{n} = -\mathbf{k}] \\
 \text{or } \iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} \, dS = \iint_{S_1} \text{curl } \mathbf{F} \cdot \mathbf{k} \, dS = \iint_{S_1} dS = S_1. \\
 \therefore S_1 = \text{area of a circle of radius 1} = \pi (1)^2 = \pi, \\
 \therefore \iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} \, dS = \pi. \quad \dots(2)
 \end{aligned}$$

Hence from (1) and (2), Stoke's theorem is verified.

Ex. 4. Apply Stoke's theorem to calculate $\int_C (4y \, dx + 2z \, dy + 6y \, dz)$, where C is the curve of intersection of $x^2 + y^2 + z^2 = 6z$ and $z = x + 3$.

Solution. We have $\int_C (4y \, dx + 2z \, dy + 6y \, dz) = \int_C \mathbf{F} \cdot d\mathbf{r}$,

where $\mathbf{F} = 4y \mathbf{i} + 2z \mathbf{j} + 6y \mathbf{k}$ and $\mathbf{r} = x \mathbf{i} + y \mathbf{j} + z \mathbf{k}$.

By Stoke's theorem $\int_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS$,

where S is the surface of the circle $x^2 + y^2 + z^2 = 6z$, $z = x + 3$ and $\mathbf{n}$ is normal to the plane $x - z + 3 = 0$.

$$\therefore \mathbf{n} = \frac{\nabla \phi}{|\nabla \phi|} = \frac{\left(\mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z} \right) \cdot (x - z + 3)}{|\nabla \phi|} = \frac{\mathbf{i} - \mathbf{k}}{\sqrt{(1+1)}} = \frac{\mathbf{i} - \mathbf{k}}{\sqrt{2}}$$

$$\text{and } \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 4y & 2z & 6y \end{vmatrix} = (6-2)\mathbf{i} - (0-0)\mathbf{j} + (0-4)\mathbf{k} = 4\mathbf{i} - 4\mathbf{k}.$$

$$\therefore (\nabla \times \mathbf{F}) \cdot \mathbf{n} = (4\mathbf{i} - 4\mathbf{k}) \cdot \frac{\mathbf{i} - \mathbf{k}}{\sqrt{2}} = \frac{4+4}{\sqrt{2}} = 4\sqrt{2}.$$

$$\text{Thus, } \int_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS = \iint_S 4\sqrt{2} \, dS = 4\sqrt{2} S = 4\sqrt{2} (\text{area of circle}).$$

Now the centre of the sphere $x^2 + y^2 + (z-3)^2 = 9$, $(0, 0, 3)$ lies on the plane $z = x + 3$. It means that the given circle is a great circle of the sphere.

$\therefore$ Radius of the circle = the radius of the sphere = 3

and so area of the circle = $\pi (3)^2 = 9\pi$.

$$\text{Therefore, } \int_C \mathbf{F} \cdot d\mathbf{r} = 4\sqrt{2}(9\pi) = 36\sqrt{2}\pi.$$

$$\text{Hence, } \int_C (4y \, dx + 2z \, dy + 6y \, dz) = 36\sqrt{2}\pi.$$

Ex. 5. Verify Stoke's theorem for the vector $\mathbf{F} = z \mathbf{i} + x \mathbf{j} + y \mathbf{k}$ taken over the half of the sphere $x^2 + y^2 + z^2 = 1^2$ lying above the xy -plane.

Solution. Here let S be the surface of the sphere $x^2 + y^2 + z^2 = 1^2$ lying above the xy -plane and let the curve C be the boundary of this surface. Obviously the curve C is a circle in the xy -plane of radius 1 and centre origin and its equations are $x^2 + y^2 = 1^2, z = 0$.

Suppose $x = \cos t, y = \sin t, z = 0, 0 \leq t < 2\pi$ are parametric equations of C .

By Stoke's theorem, we have $\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} dS$ (1)

Let us verify (1). We have

$$\begin{aligned}\oint_C \mathbf{F} \cdot d\mathbf{r} &= \oint_C (z \mathbf{i} + x \mathbf{j} + y \mathbf{k}) \cdot (dx \mathbf{i} + dy \mathbf{j} + dz \mathbf{k}) \\ &= \oint_C (z dx + x dy + y dz) = \oint_C x dy, \text{ since on } C, z = 0 \text{ and } dz = 0 \\ &= \int_0^{2\pi} \cos t \cdot \frac{dy}{dt} dt = \int_0^{2\pi} \cos t \cdot \cos t dt \\ &= \int_0^{2\pi} \cos^2 t dt = \frac{1}{2} \int_0^{2\pi} (1 + \cos 2t) dt = \frac{1}{2} \left[t + \frac{\sin 2t}{2} \right]_0^{2\pi} = \frac{1}{2} \cdot 2\pi \\ &= \pi. \quad \dots(2)\end{aligned}$$

Now let us find $\iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} dS$. We have

$$\text{curl } \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ z & x & y \end{vmatrix} = \mathbf{i} + \mathbf{j} + \mathbf{k}.$$

If $\mathbf{n}$ is a unit vector along outward drawn normal at any point (x, y, z) on the surface S i.e., the surface $\phi(x, y, z) = x^2 + y^2 + z^2 = 1$, then

$$\mathbf{n} = \frac{\nabla \phi}{|\nabla \phi|} = \frac{2x \mathbf{i} + 2y \mathbf{j} + 2z \mathbf{k}}{\sqrt{4x^2 + 4y^2 + 4z^2}} = \frac{x \mathbf{i} + y \mathbf{j} + z \mathbf{k}}{1}, \text{ since on } S, x^2 + y^2 + z^2 = 1^2.$$

$$\therefore \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} dS = \iint_S (\mathbf{i} + \mathbf{j} + \mathbf{k}) \cdot (x \mathbf{i} + y \mathbf{j} + z \mathbf{k}) dS = \iint_S (x + y + z) dS.$$

To evaluate it, we shall use spherical polar coordinates (r, θ, ϕ) . Here

$x = r \sin \theta \cos \phi, y = r \sin \theta \sin \phi, z = r \cos \theta$ and $r = 1$.

$\therefore x = \sin \theta \cos \phi, y = \sin \theta \sin \phi, z = \cos \theta$.

Also dS = an elementary area on the surface of the sphere at the point $(1, \theta, \phi)$
 $= (1 d\theta) \cdot (1 \sin \theta d\phi) = \sin \theta d\theta d\phi$.

$$\begin{aligned}\therefore \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} dS &= \int_{\theta=0}^{\pi/2} \int_{\phi=0}^{2\pi} (\sin \theta \cos \phi + \sin \theta \sin \phi + \cos \theta) \sin \theta d\theta d\phi \\ &= \int_{\theta=0}^{\pi/2} \int_{\phi=0}^{2\pi} (\sin^2 \theta \cos \phi + \sin^2 \theta \sin \phi + \cos \theta \sin \theta) d\theta d\phi \\ &= \int_{\theta=0}^{\pi/2} \left[\sin^2 \theta \sin \phi - \sin^2 \theta \cos \phi + \phi \cos \theta \sin \theta \right]_{\phi=0}^{2\pi} d\theta \\ &= \int_{\theta=0}^{\pi/2} 2\pi \cos \theta \sin \theta d\theta = 2\pi \cdot \frac{1}{2} = \pi. \quad \dots (3)\end{aligned}$$

$$= \int_{\theta=0}^{\pi/2} 2\pi \cos \theta \sin \theta d\theta = 2\pi \cdot \frac{1}{2} = \pi. \quad \dots (3)$$

From (2) and (3), we see that

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} dS.$$

This verifies Stoke's theorem.

Ex. 6. Use Stoke's theorem to evaluate $\int_C \mathbf{v} \cdot d\mathbf{r}$, where $\mathbf{v} = y^2 \mathbf{i} + xy \mathbf{j} + xz \mathbf{k}$ and C is the bounding curve of the hemisphere $x^2 + y^2 + z^2 = 9$, $z > 0$, oriented in the positive direction.

Solution. By Stoke's theorem, we have $\oint_C \mathbf{v} \cdot d\mathbf{r} = \iint_S (\text{curl } \mathbf{v}) \cdot \mathbf{n} dS$, ... (1)

where S is the surface bounded by closed curve C and $\mathbf{n}$ is the outward drawn unit normal vector to the surface S .

$$\text{Now curl } \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 & xy & xz \end{vmatrix} = (0 - 0) \mathbf{i} - (z - 0) \mathbf{j} + (y - 2y) \mathbf{k} = -z\mathbf{j} - y\mathbf{k},$$

$$\mathbf{n} = \frac{\nabla \phi}{|\nabla \phi|} = \frac{\left(\mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z} \right) (x^2 + y^2 + z^2 - 9)}{|\nabla \phi|}$$

$$= \frac{2x\mathbf{i} + 2y\mathbf{j} + 2z\mathbf{k}}{\sqrt{4x^2 + 4y^2 + 4z^2}} = \frac{x\mathbf{i} + y\mathbf{j} + z\mathbf{k}}{\sqrt{x^2 + y^2 + z^2}} = \frac{x\mathbf{i} + y\mathbf{j} + z\mathbf{k}}{3}$$

$$\text{and } (\nabla \times \mathbf{v}) \cdot \mathbf{n} = (-z\mathbf{j} - y\mathbf{k}) \cdot \frac{x\mathbf{i} + y\mathbf{j} + z\mathbf{k}}{3} = \frac{-yz - yz}{3} = \frac{-2yz}{3}.$$

$$\text{We have } \mathbf{n} \cdot \mathbf{k} dS = dx dy \quad \text{or} \quad \frac{x\mathbf{i} + y\mathbf{j} + z\mathbf{k}}{3} \cdot \mathbf{k} dS = dx dy \quad \text{or} \quad \frac{z}{3} dS = dx dy$$

$$\text{or } dS = \frac{3}{z} dx dy.$$

$$\therefore \iint_S (\nabla \times \mathbf{v}) \cdot \mathbf{n} dS = \iint \left(\frac{-2yz}{3} \right) \left(\frac{3}{z} dx dy \right) = - \iint 2y dx dy = -2 \int_0^{2\pi} \int_0^3 r \sin \theta \cdot r d\theta dr$$

$$= -2 \int_0^{2\pi} \sin \theta d\theta \int_0^3 r^2 dr = -2 \left[-\cos \theta \right]_0^{2\pi} \cdot \left[\frac{r^3}{3} \right]_0^3 = -2(-1 + 1) 9 = 0.$$

Putting this value in (1), we have $\oint_C \mathbf{v} \cdot d\mathbf{r} = 0$.

Ex. 7. Verify Stoke's theorem for the vector $\mathbf{A} = 3y \mathbf{i} - xz \mathbf{j} + yz^2 \mathbf{k}$, where S is the surface of the paraboloid $2z = x^2 + y^2$ cut by $z = 2$ and C is its boundary.

Solution. The boundary C of the surface S is the circle in the plane $z = 2$ whose equations are $x^2 + y^2 = 4$, $z = 2$. The radius of this circle is 2 and centre $(0, 0, 2)$. Suppose $x = 2 \cos t$, $y = 2 \sin t$, $z = 2$, $0 \leq t < 2\pi$ are parametric equations of C . By Stoke's theorem $\oint_C \mathbf{A} \cdot d\mathbf{r} = \iint_S (\text{curl } \mathbf{A}) \cdot \mathbf{n} dS$,

where $\mathbf{n}$ is a unit vector along outward drawn normal to the surface S .

We have $\oint_C \mathbf{A} \cdot d\mathbf{r} = \oint_C (3y \mathbf{i} - xz \mathbf{j} + yz^2 \mathbf{k}) \cdot (dx \mathbf{i} + dy \mathbf{j} + dz \mathbf{k})$

$$= \oint_C (3y dx - xz dy + yz^2 dz)$$

$$= \oint_C (3y dx - 2x dy), \text{ since on } C, z = 2 \text{ and } dz = 0$$

$$= \int_{2\pi}^0 \left(3y \frac{dx}{dt} - 2x \frac{dy}{dt} \right) dt \quad \begin{array}{l} \text{[Here surface } S \text{ lies below the curve } C. \\ \text{So the direction of } C \text{ is positive,} \\ \text{if } C \text{ is traversed in clockwise sense]} \end{array}$$

$$= - \int_0^{2\pi} [3 \times 2 \sin t \cdot (-2 \sin t) - 2 \times 2 \cos t (2 \cos t)] dt$$

$$= - \int_0^{2\pi} [-12 \sin^2 t - 8 \cos^2 t] dt = 12 \int_0^{2\pi} \sin^2 t dt + 8 \int_0^{2\pi} \cos^2 t dt$$

$$= 12 \cdot \frac{1}{2} \times 2\pi + 8 \cdot \frac{1}{2} \times 2\pi = 20\pi. \quad \dots (1)$$

Let S_1 be the plane region bounded by the circle C . If S' is the surface consisting of the surfaces S and S_1 , then S' is a closed surface. Let V be the volume bounded by S' .

By Gauss divergence theorem, we have

$$\iint_{S'} (\text{curl } \mathbf{A}) \cdot \mathbf{n} dS = \iiint_V \text{div curl } \mathbf{A} dV = 0, \text{ since } \text{div curl } \mathbf{A} = 0$$

$$\iint_S (\text{curl } \mathbf{A}) \cdot \mathbf{n} dS + \iint_{S_1} (\text{curl } \mathbf{A}) \cdot \mathbf{n} dS = 0 \quad [\because S' \text{ consists of } S \text{ and } S_1]$$

or

$$\iint_S (\text{curl } \mathbf{A}) \cdot \mathbf{n} dS = - \iint_{S_1} (\text{curl } \mathbf{A}) \cdot \mathbf{n} dS = - \iint_{S_1} (\text{curl } \mathbf{A}) \cdot \mathbf{k} dS.$$

$[\because \text{ on } S_1, \mathbf{n} = \mathbf{k}]$

Now $\text{curl } \mathbf{A} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3y & -xz & yz^2 \end{vmatrix}$

$$= \mathbf{i} \left[\frac{\partial}{\partial y} (yz^2) - \frac{\partial}{\partial z} (-xz) \right] - \mathbf{j} \left[\frac{\partial}{\partial x} (yz^2) - \frac{\partial}{\partial z} (3y) \right] + \mathbf{k} \left[\frac{\partial}{\partial x} (-xz) - \frac{\partial}{\partial y} (3y) \right]$$

$$= (z^2 + x) \mathbf{i} - (z + 3) \mathbf{k}.$$

$$\therefore \iint_S (\text{curl } \mathbf{A}) \cdot \mathbf{n} dS = - \iint_{S_1} [(z^2 + x) \mathbf{i} - (z + 3) \mathbf{k}] \cdot \mathbf{k} dS$$

$$= \iint_{S_1} (z + 3) dS = \iint_{S_1} 5 dS, \text{ since on } S_1, z = 2$$

$$= 5S_1, \text{ where } S_1 \text{ is the area of a circle of radius 2}$$

$$= 5 \cdot \pi \cdot 2^2 = 20\pi. \quad \dots (2)$$

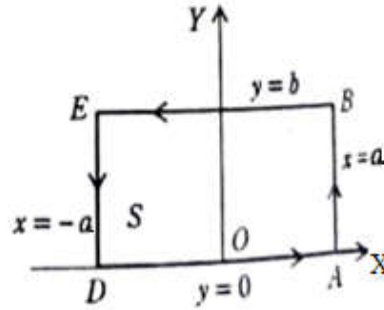
From (1) and (2), we see that $\oint_C \mathbf{A} \cdot d\mathbf{r} = \iint_S (\text{curl } \mathbf{A}) \cdot \mathbf{n} dS.$

This verifies Stoke's theorem.

Ex. 8. Verify Stoke's theorem for $\mathbf{F} = (x^2 + y^2)\mathbf{i} - 2xy\mathbf{j}$ taken round the rectangle bounded by $x = \pm a$, $y = 0$, $y = b$.

Solution. We have

$$\begin{aligned}\text{curl } \mathbf{F} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 + y^2 & -2xy & 0 \end{vmatrix} \\ &= (-2y - 2y)\mathbf{k} = -4y\mathbf{k}.\end{aligned}$$



Also $\mathbf{n} = \mathbf{k}$.

$$\begin{aligned}\therefore \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS &= \int_{y=0}^b \int_{x=-a}^a (-4y\mathbf{k}) \cdot \mathbf{k} \, dx \, dy \\ &= -4 \int_{y=0}^b \int_{x=-a}^a y \, dx \, dy = -4 \int_{y=0}^b [xy]_{x=-a}^a \, dy \\ &= -4 \int_{y=0}^b 2ay \, dy = -4 [ay^2]_0^b = -4ab^2.\end{aligned}$$

$$\begin{aligned}\text{Now } \oint_C \mathbf{F} \cdot d\mathbf{r} &= \oint_C [(x^2 + y^2)\mathbf{i} - 2xy\mathbf{j}] \cdot (dx\mathbf{i} + dy\mathbf{j}) \\ &= \oint_C [(x^2 + y^2)dx - 2xy \, dy] = \left[\int_{DA} + \int_{AB} + \int_{BE} + \int_{ED} \right] [(x^2 + y^2)dx - 2xy \, dy]\end{aligned}$$

Along DA, $y = 0$ and $dy = 0$. Along AB, $x = a$ and $dx = 0$.

Along BE, $y = b$ and $dy = 0$. Along ED, $x = -a$ and $dx = 0$.

$$\begin{aligned}\therefore \oint_C \mathbf{F} \cdot d\mathbf{r} &= \int_{x=-a}^a x^2 \, dx + \int_{y=0}^b -2ay \, dy + \int_{x=a}^{-a} (x^2 + b^2) \, dx + \int_{y=b}^0 2ay \, dy \\ &= \int_{-a}^a x^2 \, dx - \int_{-a}^a (x^2 + b^2) \, dx - 4a \int_0^b y \, dy \\ &= - \int_{-a}^a x^2 \, dx - 4a \int_0^b y \, dy = -2ab^2 - 4a \left[\frac{y^2}{2} \right]_0^b = -4ab^2.\end{aligned}$$

Thus $\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS$. Hence the theorem is verified.

Ex. 9. Evaluate the surface integral $\int_S (\text{curl } \vec{F}) \cdot \hat{n} dS$ by transforming it into a line integral, S being that part of the surface of the paraboloid $z = 1 - x^2 - y^2$, for which $z \geq 0$ and $\vec{F} = y\hat{i} + z\hat{j} + x\hat{k}$.

Solution. The boundary C of the surface S is the circle $x^2 + y^2 = 1, z = 0$ whose parametric equations are $x = \cos t, y = \sin t, z = 0; 0 \leq t \leq 2\pi$.

By Stoke's theorem, we have

$$\begin{aligned} \int_S (\text{curl } \vec{F}) \cdot \hat{n} dS &= \int_C \vec{F} \cdot d\vec{r} \\ &= \int_C (y\hat{i} + z\hat{j} + x\hat{k}) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k}) \\ &= \int_C (ydx + zdy + xdz) \\ &= \int_C (ydx + 0 + 0) \\ &= \int_0^{2\pi} \sin t (-\sin t dt) \\ &= -\frac{1}{2} \int_0^{2\pi} 2 \sin^2 t dt \\ &= -\frac{1}{2} \int_0^{2\pi} (1 - \cos 2t) dt \\ &= -\frac{1}{2} \left(t - \frac{\sin 2t}{2} \right) \Big|_0^{2\pi} \\ &= -\frac{1}{2} [(2\pi - 0) - (0 - 0)] \\ &= -\pi. \quad \text{Ans.} \end{aligned}$$

Ex. 10. Verify Stoke's theorem for the function $\vec{F} = x^2 \hat{i} + xy \hat{j}$, integrated along the rectangle, in the plane $z=0$, whose sides are along the lines $x=0, y=0, x=a$ and $y=b$.

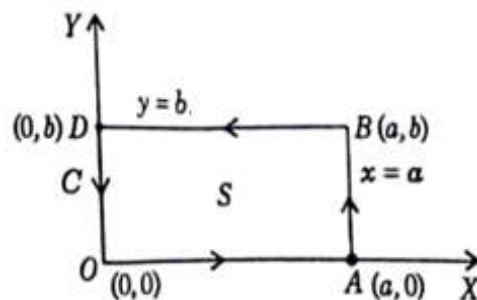
Solution. We have $\text{curl } \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 & xy & 0 \end{vmatrix} = 0\hat{i} - 0\hat{j} + \left[\frac{\partial}{\partial x}(xy) - \frac{\partial}{\partial y}(x^2) \right] \hat{k} = y\hat{k}$.

The closed curve C is the boundary of the rectangle $OABD$ traversed in anti-clockwise sense. The surface S bounded by C is the area of rectangle $OABD$.

Also $\vec{n}$ = unit normal vector to $S = \hat{k}$.

By Stoke's theorem, we have

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\text{curl } \vec{F}) \cdot \vec{n} dS.$$



Let us verify it. We have

$$\begin{aligned}\iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS &= \int_{y=0}^b \int_{x=0}^a (y \mathbf{k}) \cdot \mathbf{k} \, dx \, dy \\ &= \int_{y=0}^b \int_{x=0}^a y \, dx \, dy = \int_{x=0}^a \left[\frac{y^2}{2} \right]_{y=0}^b dx = \frac{b^2}{2} \int_0^a dx = \frac{ab^2}{2}. \quad \dots (1)\end{aligned}$$

$$\begin{aligned}\text{Also } \oint_C \mathbf{F} \cdot d\mathbf{r} &= \oint_C (x^2 \mathbf{i} + xy \mathbf{j}) \cdot (dx \mathbf{i} + dy \mathbf{j}) = \oint_C (x^2 dx + xy dy) \\ &= \int_{OA} (x^2 dx + xy dy) + \int_{AB} (x^2 dx + xy dy) + \int_{BD} (x^2 dx + xy dy) + \int_{DO} (x^2 dx + xy dy)\end{aligned}$$

Along OA, $y = 0$ and $dy = 0$, x varies from 0 to a ; along AB, $x = a$, $dx = 0$, y varies from 0 to b ; along BD, $y = b$, $dy = 0$, x varies from a to 0; and along DO, $x = 0$, $dx = 0$, y varies from b to 0.

$$\begin{aligned}\therefore \oint_C \mathbf{F} \cdot d\mathbf{r} &= \int_0^a x^2 dx + \int_0^b ay dy + \int_a^0 x^2 dx + \int_b^0 0 dy \\ &= \left[\frac{x^3}{3} \right]_0^a + a \left[\frac{y^2}{2} \right]_0^b + \left[\frac{x^3}{3} \right]_a^0 + 0 = \frac{a^3}{3} + \frac{ab^2}{2} - \frac{a^3}{3} = \frac{ab^2}{2}. \quad \dots (2)\end{aligned}$$

From (1) and (2), we see that $\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS$.

This verifies Stoke's theorem.

Ex. 11. Evaluate $\oint_C \vec{F} \cdot d\vec{r}$ by Stoke's theorem, where $\vec{F} = y^2\hat{i} + x^2\hat{j} - (x+z)\hat{k}$ and C is the boundary of the triangle with vertices at (0, 0, 0), (1, 0, 0) and (1, 1, 0).

Solution. Try yourself.

Ex. 12. Verify Stoke's theorem for $\vec{F} = (x+y)\hat{i} + (2x-z)\hat{j} + (y+z)\hat{k}$ for the surface of a triangular lamina with vertices (2, 0, 0), (0, 3, 0) and (0, 0, 6).

Solution. Try yourself.

Ex. 13. Use the Stoke's theorem to evaluate

$$\int_C [(x+2y)dx + (x-z)dy + (y-z)dz],$$

where C is the boundary of the triangle with vertices (2, 0, 0), (0, 3, 0) and (0, 0, 6) oriented in the anti-clockwise direction.

Solution. Try yourself.